

Milestone 1: Team Formation

[Process Book Link](#)

[Website Link](#)

Team Members:

Dani Ebaseh-Onofa

debasehonofa@college.harvard.edu

Oscar Boccelli

oboccelli@college.harvard.edu

Garland Catlette

gcatlette@college.harvard.edu

Milestone 2: Team Agreement

Roles and Responsibilities:

- Coordinator: Oscar (Team Meeting Scheduling/Weekly Check Ins)
- Storyteller: Dani (Creating Narratives/Documenting Story)
- Researcher: Garland (Sourcing Data/Information)
- Coder: All

Communication Protocols:

- Text Message/Zoom if necessary

Working Arrangements:

- Currier Dining Hall/Quad
- Meet Twice a Week To Progress but Flexible

Work Allocation:

- Context Dependent/Flexible
- Tasks divided as indicated above

Accountability:

- Open Communication
- Come to our meetings prepared with at least one insight or something to discuss

Signed Off:

Oscar Boccelli

Dani Ebaseh-Onofa

Garland Catlette

Milestone 3: Project Proposal

Basic Information

- **Project Title:** A Spatial Analysis of Food Deserts & Mental Health Outcomes
 - Mental Health
 - Drug Usage **
 - Obesity **
 - Life Expectancy **
- **Team Name:** TEAM DOG COB

Abstract

In this project, we seek to explore the implications of food deserts on several health outcomes, including diabetes, drug use, depression, obesity, etc. We will reference data from the US Department of Agriculture (USDA), more specifically, the Food Access Research Atlas, Drug Use Data from the National Survey on Drug Use and Health (NSDUH), and disease prevalence data from the Center for Disease Control (CDC). We've linked the data sources below. We plan to convey our narrative by constructing a storyboard with dashboards that feature interactive maps, graphs, and charts depicting the spatial prevalence of food deserts and associated health outcomes.

Background and Motivation

Given our team's collective experiences with food deserts, we wanted to dive deeper into their impacts that are not typically studied, namely the effects of food deserts on individual and collective mental health. Oscar has had prior experience working at the EMPOWER Mental Health for All Lab at Harvard Medical School, where we saw firsthand how pervasive the negative effects of mental health disorders are in disenfranchised communities. Dani has prior research experience in public health in topics ranging from social determinants of health and complications due to obesity to the opioid epidemic. Garland has conducted prior research experience in the severity of food deserts across the United States to illustrate the underlying connections to broader socioeconomic disparities. Ultimately, we want to bring these previous experiences together to help convey the importance of highlighting how food deserts and mental health work in tandem.

Data

[Food Desert Data](#)

- Overviews food access geographically across the USA

[*Drug Use Data](#)

- Overviews drug use data geographically throughout the USA

[Health Data](#)

- Overviews the prevalence of disease throughout the USA

In regards to preprocessing, we plan to handle any missing or null values in our data sets using imputation or removal depending on the type of missingness we encounter. We will also correct any data imbalances by location or time that we notice during the data exploration phase of our project.

Milestone 4: Map

Potential Target Audiences

- Communities in food deserts (potentially community activists)
- Urban planners, city officials/policymakers
- Food policy councils
- USDA, HHS representatives
- Public health researchers
- Mental health

Final Target Audience: Policymakers working in Food Security & Mental Health

Our audience is aware of the prevalence of food deserts as an issue, but they lack the empirical evidence substantiating the consequences of the issue. They are interested in fostering equity in their respective communities by passing certain policies and regulations, and are looking for the data to support their decisionmaking. Our audience has a high level understanding of data visualization literacy, i.e. bar charts, pie charts, line graphs etc, therefore we will construct visualizations that are digestible for them. We will present details/visuals at both a high level and granular level, we'd begin with the high level to communicate in a broad sense what the prevalence of the issue is and get more granular as we aim to target specific policymakers operating in certain areas.

Potential Audience Data Questions

1. In which areas of the United States are food deserts most prevalent?
2. Where is depression most prevalent in the US?
3. What is the correlation between food deserts prevalence and depression prevalence?
4. What are some concrete things we can do in terms of nutrition to address the mental health issues of our community?
5. Which aspects of mental health are most impacted by lack of food and/or inadequate nutrition?
6. What does geographic prevalence look like for drug usage in the US?
7. What is the relationship between life expectancy and areas where food deserts are prevalent?

8. Has the prevalence of food deserts changed in recent years, and how does this relate to trends in mental health disorders?
9. What do suicide rates look like in areas where food deserts are prevalent?
10. What is the relationship between food deserts and the rate of obesity?

Data Description

From the Food Access Research Atlas:

- State County Data (Entire USA)
 - Poverty Rate (Quantitative)
 - Median Family Income (Quantitative)
 - Demographics Data (Categorical)
 - SNAP Benefits (Quantitative)
 - Relative Distance (1 mile, 10 miles, beyond 20 miles, etc.) (Quantitative)

From Drug Use Data:

- State County Data (Entire USA)
 - Marijuana Usage %
 - Cocaine Use %
 - Methamphetamine Use %
 - Binge Alcohol %
 - Cigarette %
 - Tobacco %
 - Major Depressive Episode %

From Health Data:

- State County Data (Entire USA)
 - Population (Quantitative)
 - Lack of health insurance prevalence %
 - High blood pressure prevalence %
 - Asthma prevalence %
 - Coronary heart disease prevalence %
 - Diabetes prevalence %

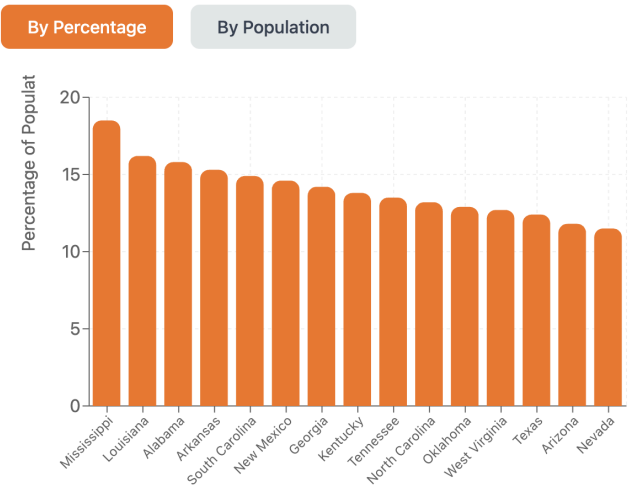
Individual Visualizations

Oscar:

Question 1: In which areas of the United States are food deserts most prevalent?

Food Deserts in the United States

States with highest prevalence of low-income, low-access areas to healthy food



<https://claude.ai/public/artifacts/22a084d5-8c0e-459d-bada-b152806ecfb2>

Question 2: Where is depression most prevalent in the US?

Depression Prevalence in the United States

Percentage of adults reporting depression by state (CDC data)



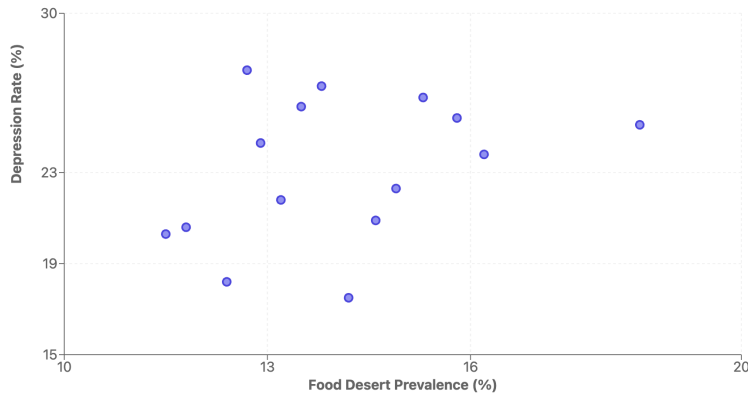
<https://claude.ai/public/artifacts/fe498c4e-2a0d-42aa-8c17-01779e146fe9>

Question 3: What is the correlation between food deserts prevalence and depression prevalence?

Exploring the relationship between food access and mental health

Scatter Plot

Side-by-Side



Correlation Analysis:

$r = 0.339$

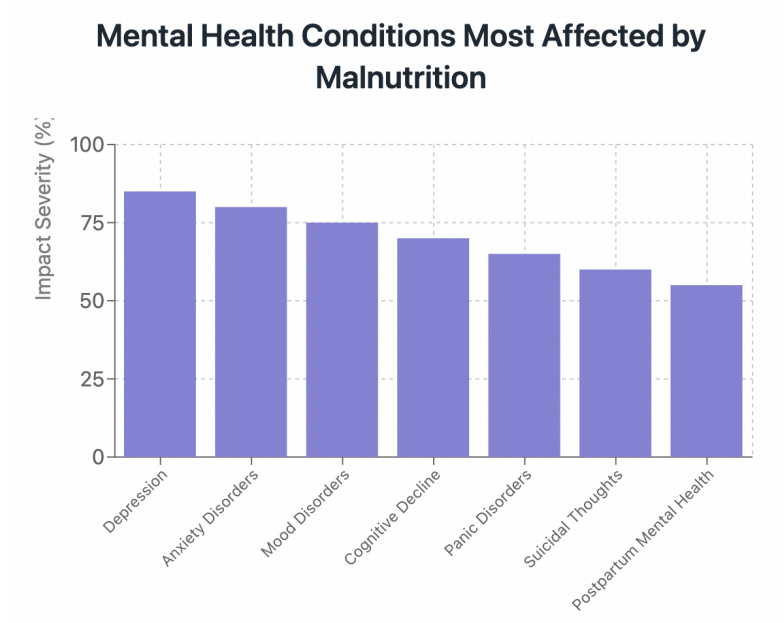
<https://claude.ai/public/artifacts/f96f0896-fc4d-4c93-8bc2-ebb5063f37fa>

Reflection:

The questions addressed in my sketches are more granular than what I had asked in my original questions. Rather than looking at Food Desert data on aggregate, my sketches hone in on specific states and localize the visualization to showing the most prevalent states/areas, which fails to reflect the gap between states where Food Deserts are more prevalent relative to those where they are less relevant. Questions that rely on geographic prevalence create more insightful visualizations relative to a correlation because correlations aren't super digestible for the average user, and ultimately we are aiming to craft visualizations that serve policymakers. We decided to stay with the original questions because we felt that they were the most insightful and relevant to our objective of creating both high level and granular visualizations.

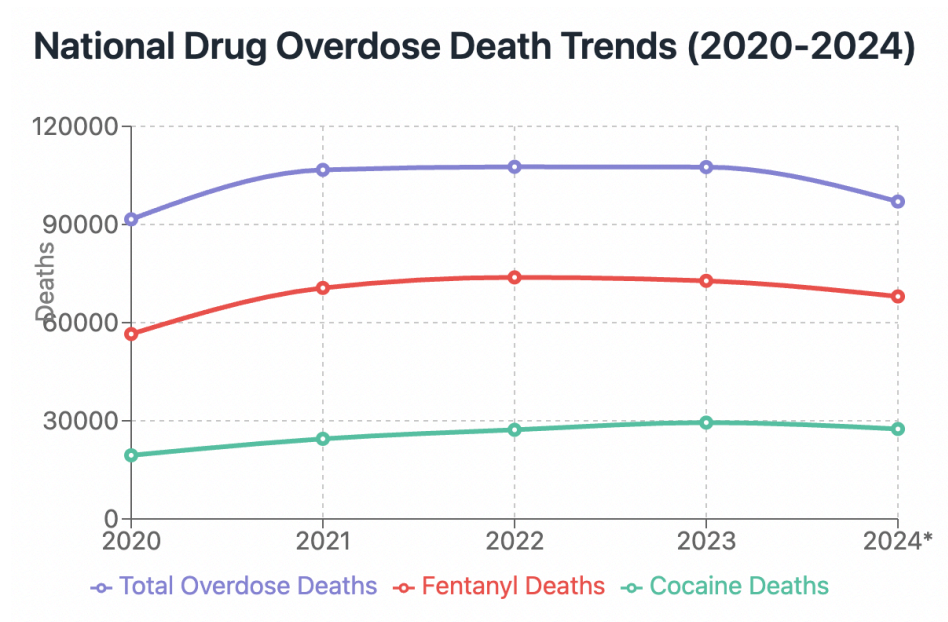
Dani:

Question 5: Which aspects of mental health are most impacted by lack of food and/or inadequate nutrition?



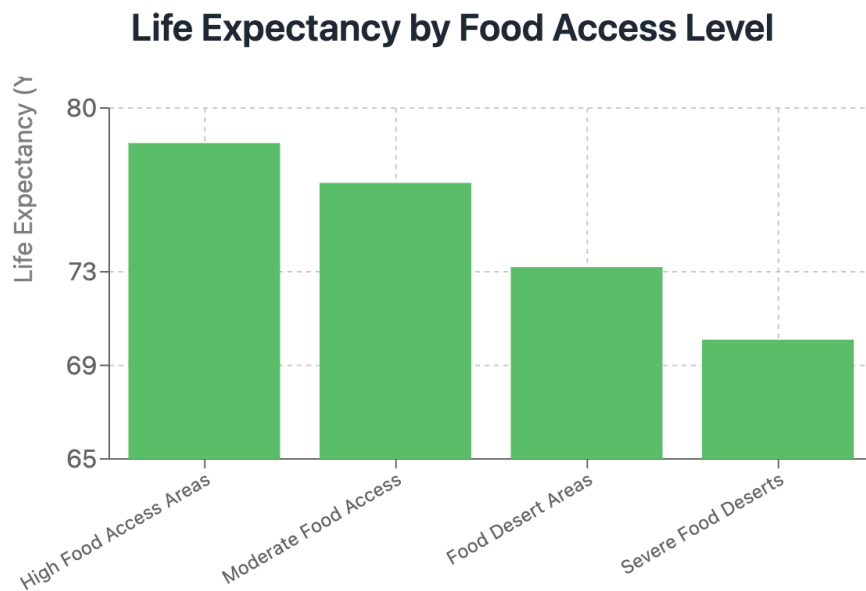
<https://claude.ai/public/artifacts/96bf476f-f646-4b68-b843-9088a7a6d4e2>

Question 6: What does geographic prevalence look like for drug usage in the U.S.?



<https://claude.ai/public/artifacts/b78281df-f5bc-4902-a5bc-b74a68a64881>

Question 7: What is the relationship between life expectancy and areas where food deserts are prevalent?



<https://claude.ai/public/artifacts/9f13515c-ef71-4fa7-8304-9f255e680842>

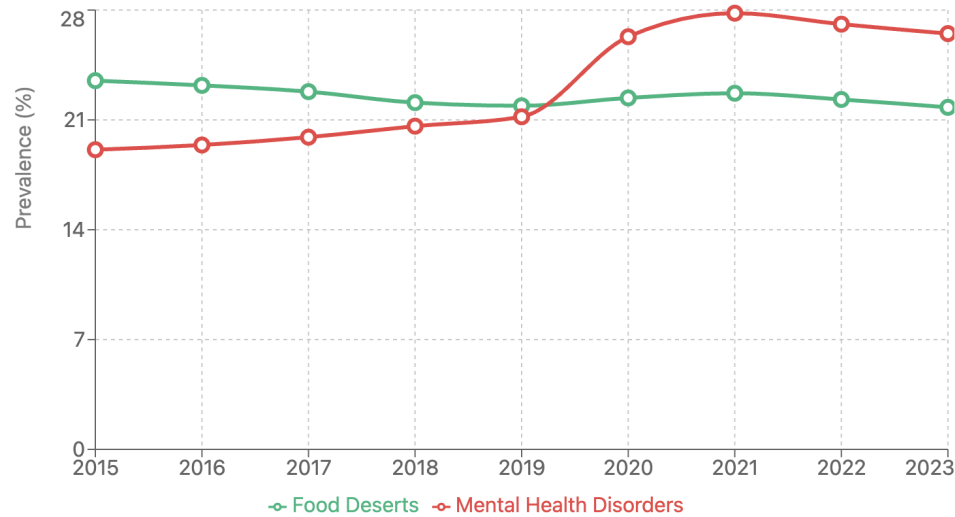
Reflection:

My sketches relate to our questions regarding which aspects of mental health are most impacted by inadequate nutrition, the geographic prevalence of drug usage in the U.S., and the relationship between life expectancy and areas where food deserts are most prevalent. The questions addressed in my sketches are similar to the original questions, but they go beyond what my team and I originally envisioned when writing the questions by including more complexity in their analysis. For instance, the sketch relating to the geographic prevalence of drug usage in the U.S. is interesting because of the breakdown of overdose deaths from cocaine and fentanyl over time which is a unique way of visualizing a seemingly simple question while including a large portion of data. I would categorize this sketch and question as better than the others because of its complexity and granularity.

Garland:

Question 8: Has the prevalence of food deserts changed in recent years, and how does this relate to trends in mental health disorders?

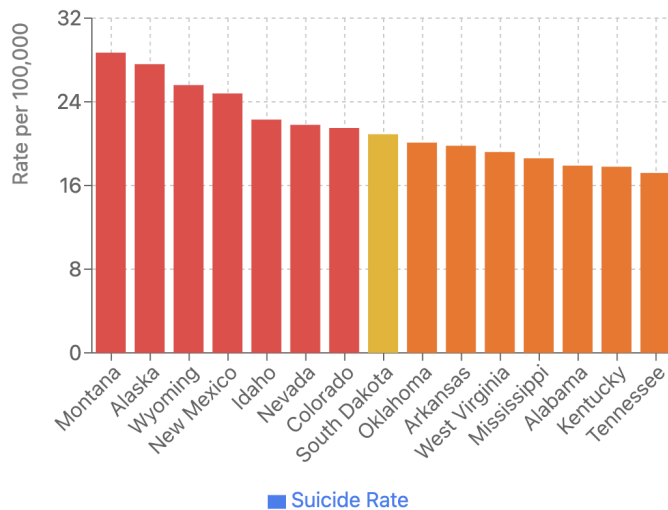
Food Desert vs Mental Health Prevalence Over Time



<https://claude.ai/chat/2bdb8f96-f8f8-4056-b13b-cad42e53ee6e>

Question 9: What do suicide rates look like in areas where food deserts are prevalent?

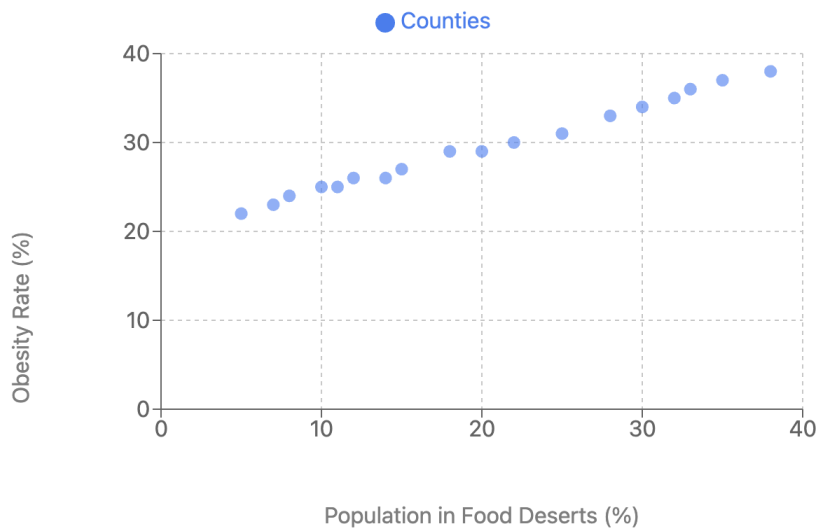
Top 15 States by Suicide Rate



<https://claude.ai/chat/2ca12c55-8b3a-4f57-8911-62456cf2cc0c>

Question 10: What is the relationship between food deserts and the rate of obesity?

Correlation Between Food Desert Prevalence and Obesity



<https://claude.ai/chat/8c7cb92f-f3eb-4676-b4fa-d69404de13a0>

Reflection:

The questions addressed in my sketches align closely with the team's original questions but demonstrate important refinements in scope and specificity. While the team originally asked about the changing prevalence of food deserts and their relationship to mental health trends over time, my first sketch directly visualized this temporal relationship, making it more concrete and answerable with available data. The second sketch maintained the team's geographic focus on suicide rates but shifted from examining "areas where food deserts are prevalent" to simply ranking states by suicide rate, which is more straightforward but loses the direct connection to food desert prevalence that made the original question compelling. The third sketch stayed true to the team's obesity question but transformed it into a correlation analysis that provides clearer visual evidence of the relationship. Overall, the original questions were stronger in their integrative approach—explicitly linking food deserts to health outcomes—whereas my sketches sometimes sacrificed this connection for visual clarity and data availability. The temporal mental health question proved most successful because it preserved both variables in a single visualization, while the suicide rate sketch would have been stronger had it maintained the geographic overlay with food desert data as originally envisioned.

Milestone 5: Data

Our final two datasets, both of which are in CSV format, are the [USDA Food Access Research Atlas](#), detailing information on food access geographically across the United States and the [Census Tract Data](#) from the U.S. Department of Health and Human Services, detailing information on disease prevalence geographically across the United States. We decided against using the [Drug Use Data](#) from the National Survey on Drug Use and Health because the location data is encoded more vaguely than the other two datasets which have a 1:1 encoding of State-County location data.

We have included our data inventories for the two datasets on separate documents for cleanliness. The links can be found below.

[Food Access Data Inventory](#)

[Health Data Inventory](#)

```
1 df = pd.read_csv("PLACES__Census_Tract_Data__GIS_Friendly_Format__2024_release.csv")
2 df.describe()
```

Visualize

We used the describe function on pandas to get statistical summaries for our data. The summaries are included in the links below.

[Food Access Data Summary](#)

[Health Data Summary](#)

Data Exploration

Food Access Data:

The list below shows the columns that had missing values along with the number of missing values in each column.

- NUMGQTRS: 25
- PCTGQTRS: 25
- PovertyRate: 3
- MedianFamilyIncome: 748
- LAPOP1_10: 29957
- LAPOP05_10: 14540
- LAPOP1_20: 35914
- LALOWI1_10: 29957
- LALOWI05_10: 14540
- LALOWI1_20: 35914
- lapophalf: 4568
- lapophalfshare: 4568
- lalowihalf: 4568
- lalowihalfshare: 4568
- lakidshalf: 4568
- lakidshalfshare: 4568
- laseniorshalf: 4568
- laseniorshalfshare: 4568
- lawwhitehalf: 4568
- lawwhitehalfshare: 4568
- lablackhalf: 4568
- lablackhalfshare: 4568
- laasianhalf: 4568
- laasianhalfshare: 4568
- lanhopihalf: 4568
- lanhopihalfshare: 4568
- laaianhalf: 4568
- laaianhalfshare: 4568
- laomultirhalf: 4568
- laomultirhalfshare: 4568
- lahisphalf: 4568
- lahisphalfshare: 4568
- lahunvhalf: 4568
- lahunvhalfshare: 4562
- lasnaphalf: 4568

- lasnaphalfshare: 4562
- lapop1: 19989
- lapop1share: 19989
- lalowil: 19989
- lalowilshare: 19989
- lakids1: 19989
- lakids1share: 19989
- laseniors1: 19989
- laseniors1share: 19989
- lawwhite1: 19989
- lawwhite1share: 19989
- lablack1: 19989
- lablack1share: 19989
- laasian1: 19989
- laasian1share: 19989
- lanhopi1: 19989
- lanhopi1share: 19989
- laaian1: 19989
- laaian1share: 19989
- laomultir1: 19989
- laomultir1share: 19989
- lahisp1: 19989
- lahisp1share: 19989
- lahunv1: 19989
- lahunv1share: 19966
- lasnap1: 19989
- lasnap1share: 19966
- lapop10: 64765
- lapop10share: 64765
- lalowil0: 64765
- lalowil0share: 64765
- lakids10: 64765
- lakids10share: 64765
- laseniors10: 64765
- laseniors10share: 64765
- lawwhite10: 64765
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- laomultir10: 64765
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- lahunv10: 64765
- lahunv10share: 64666
- lasnap10: 64765
- lasnap10share: 64666
- lapop20: 71025
- lapop20share: 71025
- lalowi20: 71025
- lalowi20share: 71025
- lakids20: 71025
- lakids20share: 71025
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- laomultir20: 71025
- laomultir20share: 71025
- lahisp20: 71025
- lahisp20share: 71025
- lahunv20: 71025
- lahunv20share: 70920
- lasnap20: 71025
- lasnap20share: 70920
- TractLOWI: 4

- TractKids: 4
- TractSeniors: 4
- TractWhite: 4
- TractBlack: 4
- TractAsian: 4
- TractNHOPI: 4
- TractAIAN: 4
- TractOMultir: 4
- TractHispanic: 4
- TractHUNV: 4
- TractSNAP: 4

Health Data

The list below shows the columns that had missing values along with the number of missing values in each column.

- BPHIGH_CrudePrev: 5077
- BPHIGH_Crude95CI: 5077
- BPMED_CrudePrev: 5077
- BPMED_Crude95CI: 5077
- CHOLSCREEN_CrudePrev: 5077
- CHOLSCREEN_Crude95CI: 5077
- COLON_SCREEN_CrudePrev: 3
- COLON_SCREEN_Crude95CI: 3
- HIGHCHOL_CrudePrev: 5077
- HIGHCHOL_Crude95CI: 5077
- MAMMOUSE_CrudePrev: 42
- MAMMOUSE_Crude95CI: 42
- TEETHLOST_CrudePrev: 34
- TEETHLOST_Crude95CI: 34
- ISOLATION_CrudePrev: 19635
- ISOLATION_Crude95CI: 19635
- FOODSTAMP_CrudePrev: 19635
- FOODSTAMP_Crude95CI: 19635
- FOODINSECU_CrudePrev: 19635
- FOODINSECU_Crude95CI: 19635
- HOUSINSECU_CrudePrev: 19635
- HOUSINSECU_Crude95CI: 19635
- SHUTUTILITY_CrudePrev: 19635
- SHUTUTILITY_Crude95CI: 19635
- LACKTRPT_CrudePrev: 19635

- LACKTRPT_Crude95CI: 19635
- EMOTIONSPT_CrudePrev: 19635
- EMOTIONSPT_Crude95CI: 19635

Data Cleaning

Food Atlas Data

```
1 threshold = 30
2 rows_to_drop = food_access_df[cols_with_missing].isnull().sum(axis=1) < threshold
3 food_access_df = food_access_df.loc[~rows_to_drop].copy()
```

- We removed rows for which the columns had a small proportion of missing values (<30 missing values)

```
1 county_col = 'County'
2 state_col = 'State'
3 cols = cols_with_missing
4
5 imputed_df = food_access_df.copy()
6
7 county_means = imputed_df.groupby(county_col)[cols].transform('mean')
8 imputed_df[cols] = imputed_df[cols].fillna(county_means)
9
10 if state_col in imputed_df.columns:
11     state_means = imputed_df.groupby(state_col)[cols].transform('mean')
12     imputed_df[cols] = imputed_df[cols].fillna(state_means)
13
14 imputed_df[cols] = imputed_df[cols].fillna(imputed_df[cols].mean())
15
16 print(imputed_df.isnull().sum())
```

- We imputed by mean by county for columns that had large proportions of missing values to still maintain our data and not needlessly remove large sets of observations
- There were no duplicates:

```
Number of duplicate rows (exact matches): 0
Total rows involved in duplicates: 0
```

✓ No exact duplicate rows found!

```
=====
NOTE: Individual column duplicates are EXPECTED
(e.g., observation values can repeat across rows)
=====
```

This script only flags rows where ALL columns are identical.

```
=====
REMOVING DUPLICATES:
=====
```

```
Rows after removing duplicates: 72533
Removed 0 duplicate rows
```

```
dhcp-10-250-161-252:CS 171 oscarboccelli$
```

Health Data

- We removed columns with missing values (prevalence of arthritis, high blood pressure, cancer, asthma, heart disease, cholesterol, dentist visits, mammography, stroke, and teeth loss), because there are several other outcomes that we are interested in from the health data set that don't have missing values, i.e. Obesity, Short Sleep Duration, etc. using the same strategy as above
- We also removed confidence interval columns due to their lack of interpretability for the sake of our creation of a visualization

```
Number of duplicate rows (exact matches): 0
Total rows involved in duplicates: 0

✓ No exact duplicate rows found!

=====
NOTE: Individual column duplicates are EXPECTED
(e.g., observation values can repeat across rows)
=====
This script only flags rows where ALL columns are identical.

=====
REMOVING DUPLICATES:
=====
Rows after removing duplicates: 83522
Removed 0 duplicate rows
○ dhcp-10-250-161-252:CS 171 oscarboccelli$ █
```

Reflection

In comparison to our original vision, our actual data is thorough enough to support our initial goal of deriving and illustrating insights pertaining to the correlation between food insecurity and mental health conditions. Furthermore, both of our data sources contain observations for counties in each state, allowing us to utilize county-specific data across various regions in the United States to build the credibility of our conclusions. With this, we can answer our initial question about where depression is most prevalent in the United States. Additionally, we can answer our initial question about the relationship between obesity and food deserts.

However, with our current data, we are unable to describe the correlation between food deserts and life expectancy. Secondly, we are unable to specify which aspects of mental health are exacerbated by food insecurity. Given that our data contains columns with missing values, we utilized county averages that could potentially misrepresent trends, as averages are sensitive to outliers. Another apparent yet reparable issue regarding our data quality would entail the transformation of numerous relevant categorical values within our data to build quantitative models.

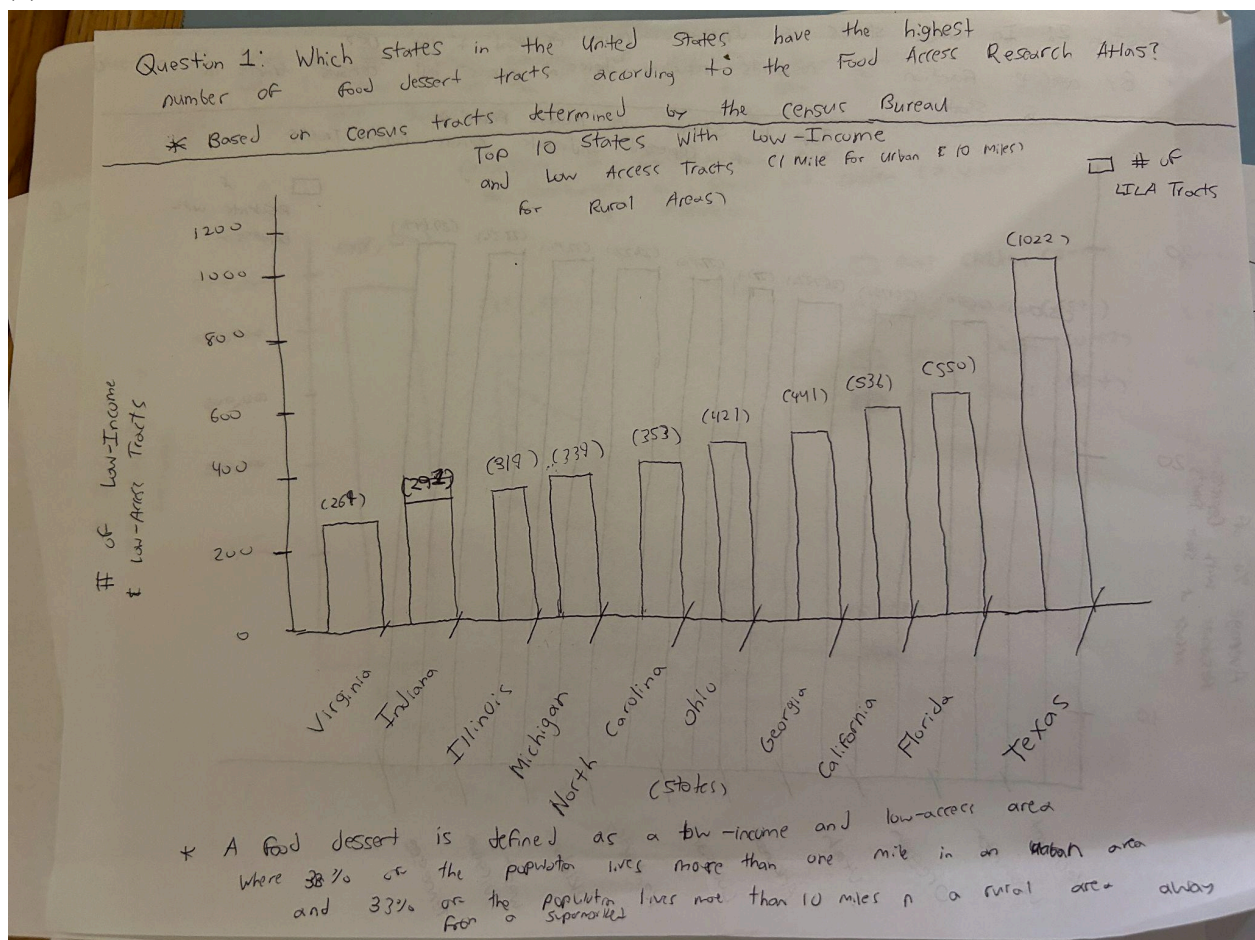
[Updated Food Access Data](#)

[Updated Aggregate Health Data Inventory](#)

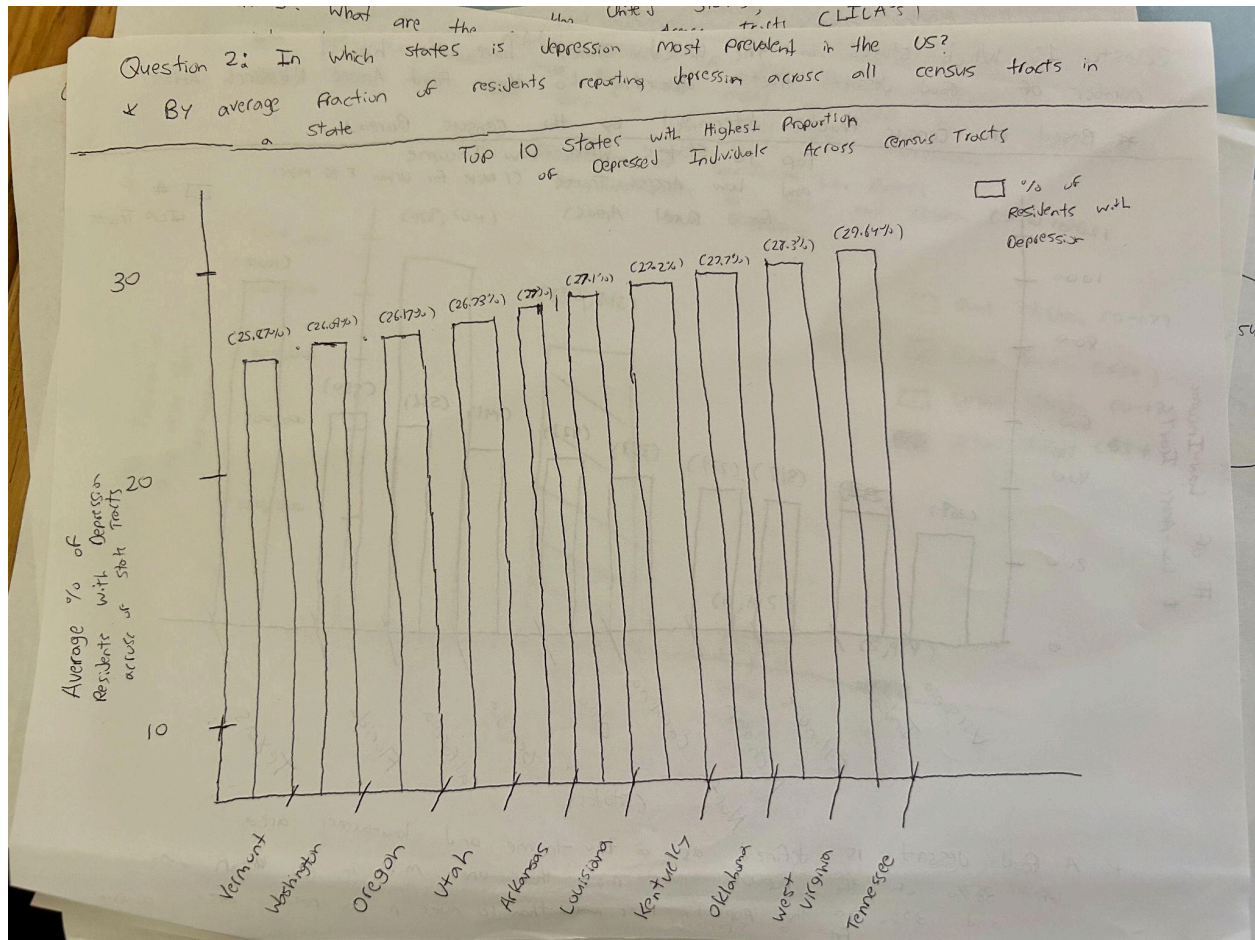
Milestone 6: Sketch

Oscar's Sketches

(1)



(2)



(3)

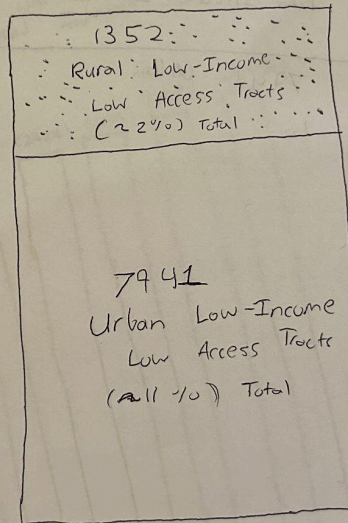
Question 3: Of the census tracts in the United States, what are the relative number of Urban/Rural Low-Income, Low-Access Tracts (LILAs)?

Urban/Rural LILA's Tracts
in the United States

72,531 Census Tracts

☒ Rural LILA's

☐ Urban LILA's



(4)

Question 9: What is the number of individuals who are more than 10 miles away from a supermarket by race?

O = # Individuals
> 10 miles
from Supermarket

4,216,545

White

383,417

Hispanic

368,531

Black

298,444

American
Indian

235,294

Mixed

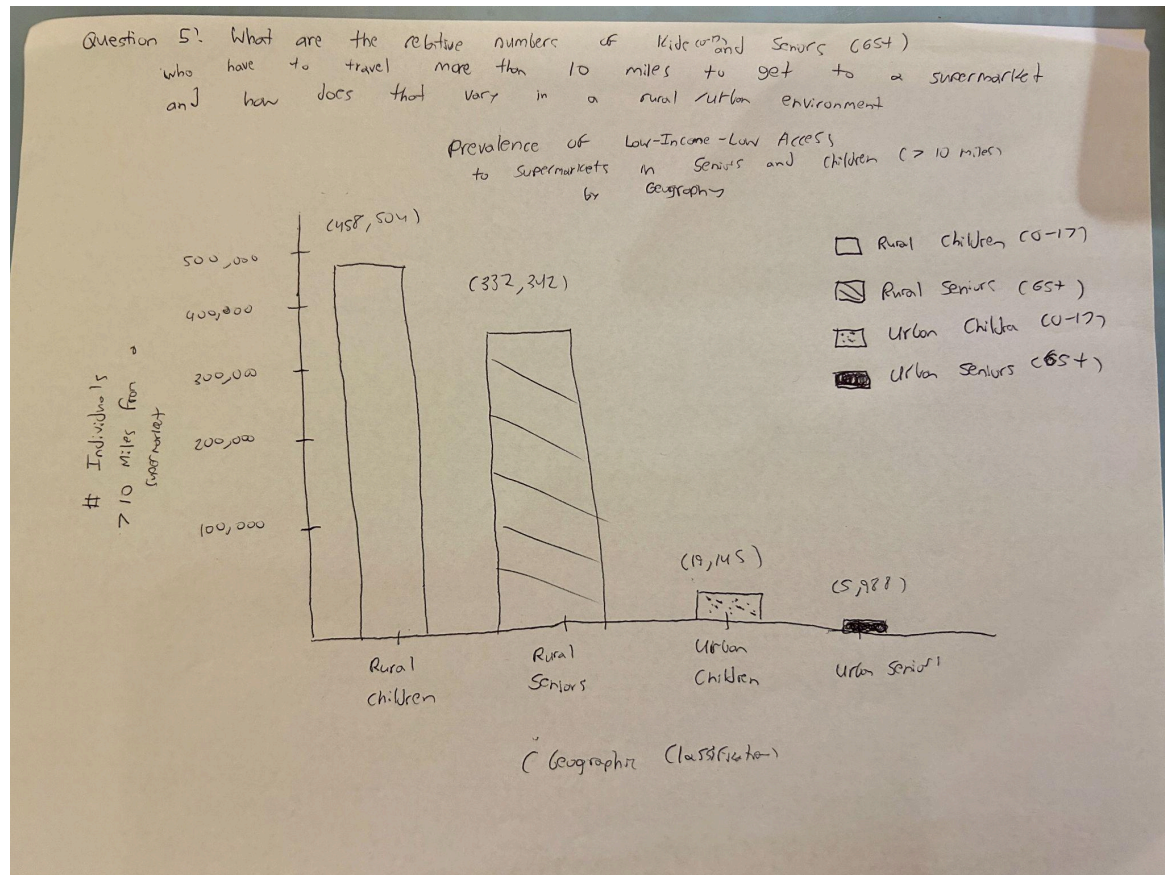
21,789

Asian

4062

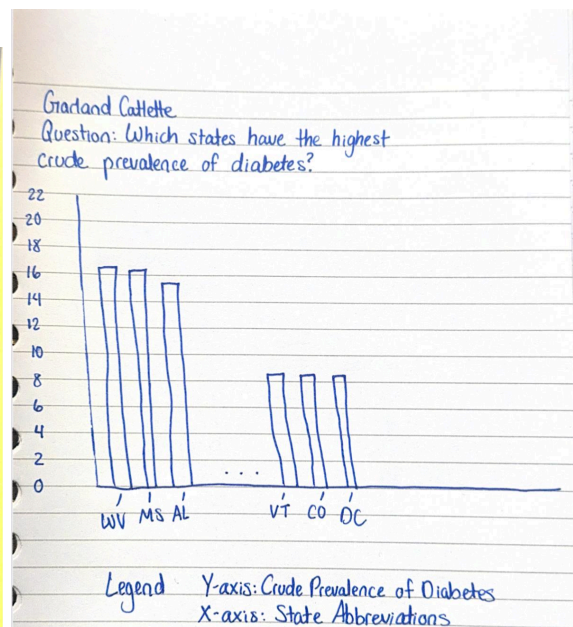
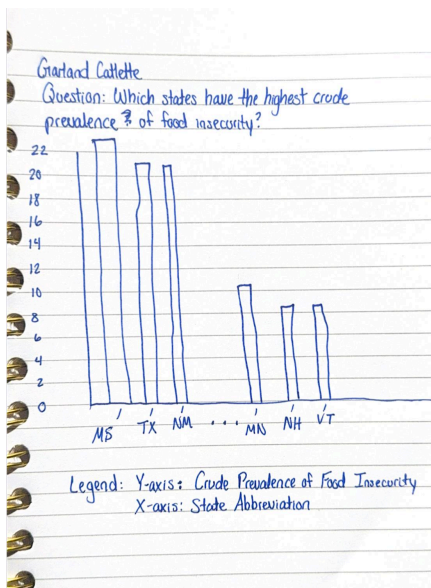
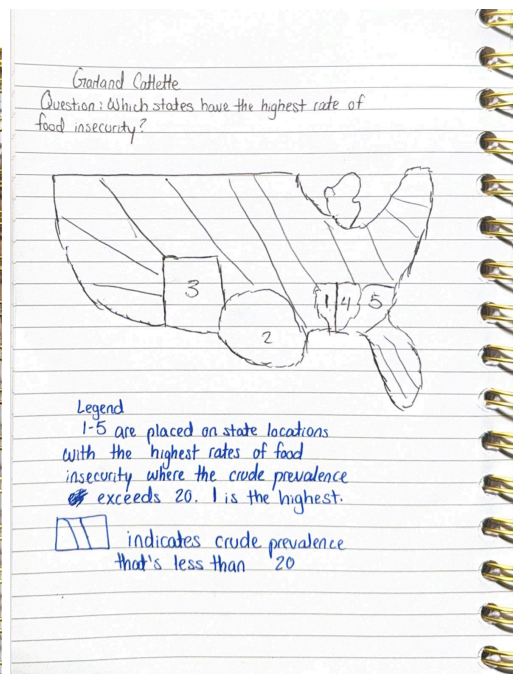
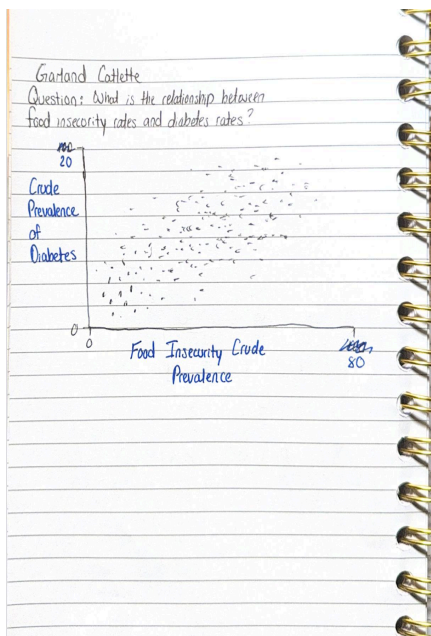
Native

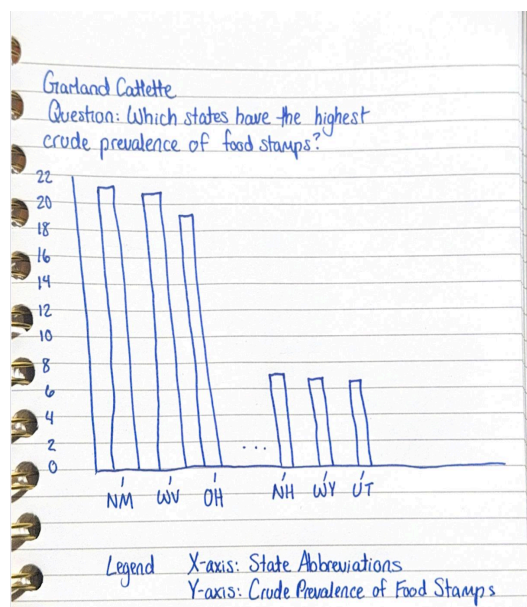
(5)



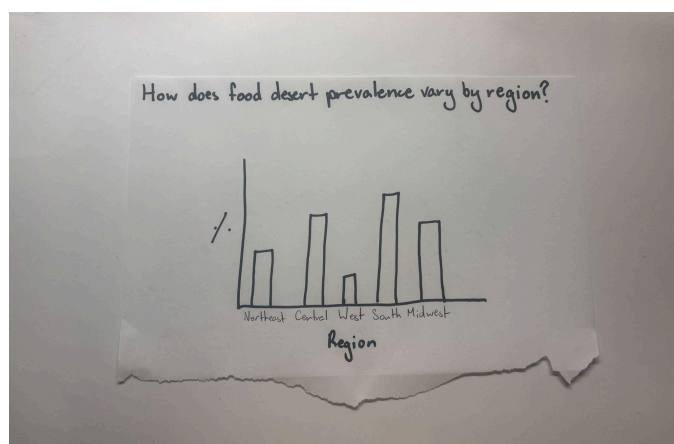
Garland's Sketches

(6 Scatterplot, 7 Map, 8 Food Insecurity Bar Graph, 9 Diabetes Bar Graph, 10 Food Stamps Bar Graph)

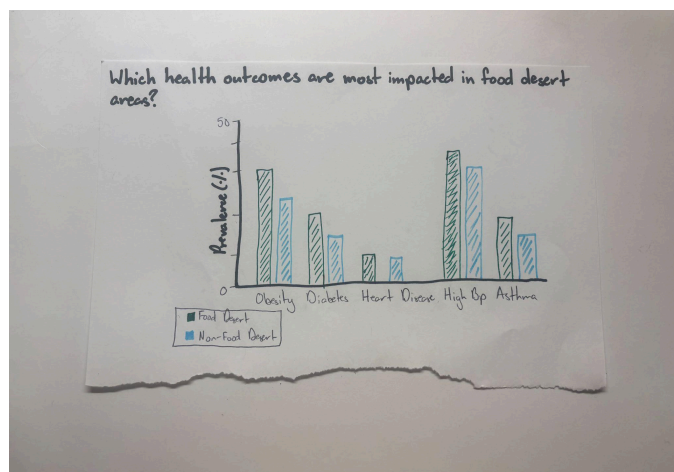




Dani

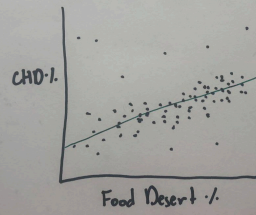


11



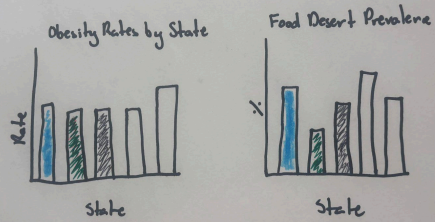
12

What is the correlation between prevalence of food deserts and prevalence of coronary heart disease?
 CHD % vs. Food Desert % (County)



13

What is the relationship between food deserts and the rate of obesity?



14

In which areas are food deserts most prevalent in the US?



15

Milestone 7: Decide

Question IDs

1. Which states in the US have the highest number of US food desert tracts according to the Food Access Research Atlas?
2. In which states is depression most prevalent in the US?
3. Of the Census Tracts in the United States what are the relative number of urban and rural low-income & low-access tracts (LILA's)
4. What is the number of individuals who are more than 10 miles away from a supermarket by race?
5. What are the relative numbers of kids (0-17) and seniors (65+) who have to travel more than 10 miles to get to a supermarket and how does that vary in a rural/urban environment?
6. How does food desert prevalence vary by region?
7. Which health outcomes are most impacted in food desert areas?
8. What is the correlation between the prevalence of food deserts and prevalence of coronary heart disease?
9. What is the relationship between food deserts and the rate of obesity?
10. In which areas are food deserts most prevalent in the US?
11. Which states have the highest food insecurity rates?
12. Which states have the highest crude prevalence of food stamps?
13. Which states have the highest crude prevalence of diabetes?
14. Which states have the highest crude prevalence of food insecurity?
15. Is there a correlation between the crude prevalence of diabetes and the crude prevalence of food insecurity?

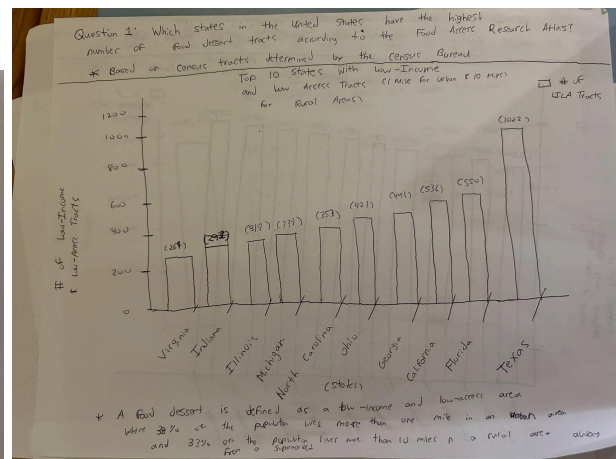
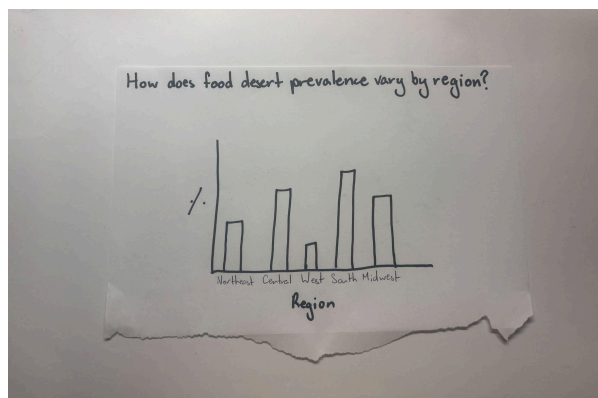
Voting Process

Sketch ID	Question ID	Author	Number of Votes
1, 11	1, 6	OB, DO	3
2	2	OB	0
3	3	OB	0
4	4	OB	2
5	5	OB	2
6	15	GC	1
10	12	GC	2

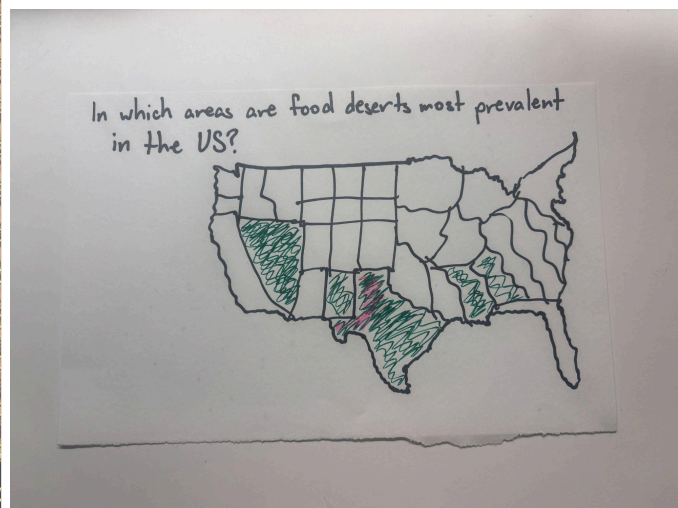
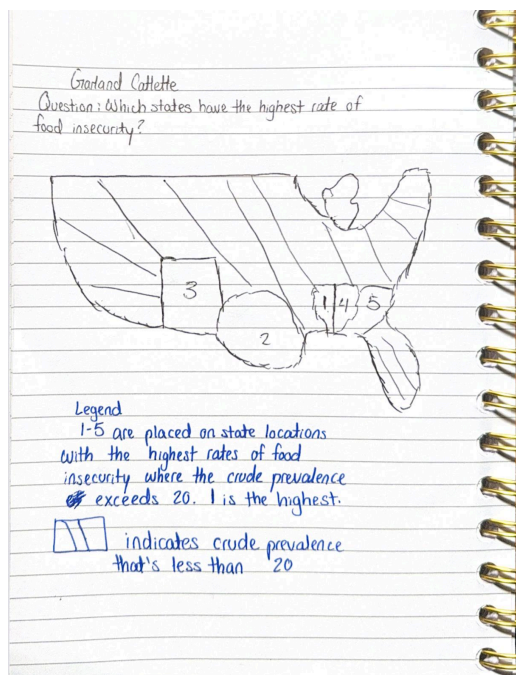
8	14	GC	0
9	13	GC	0
7, 15	10, 11	GC, DO	3
12	7	DO	2
13	8	DO	0
14	9	DO	0

Selected Sketches

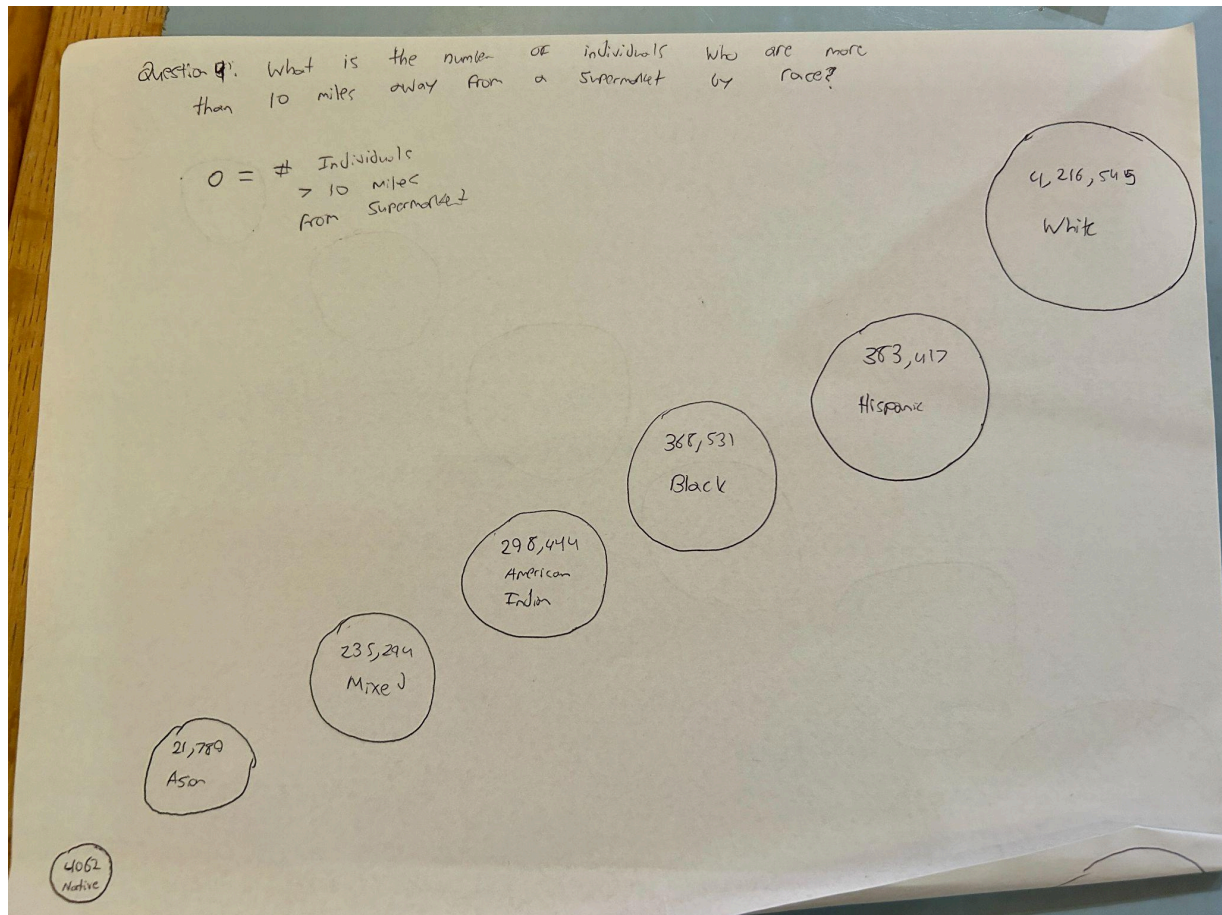
Sketch 1/11:



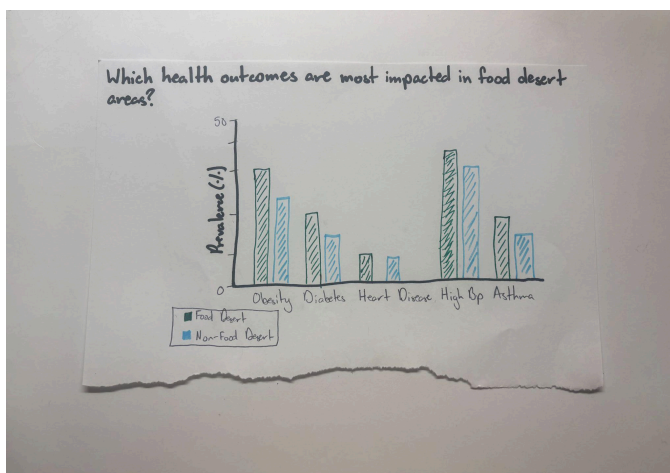
Sketch 7/15:



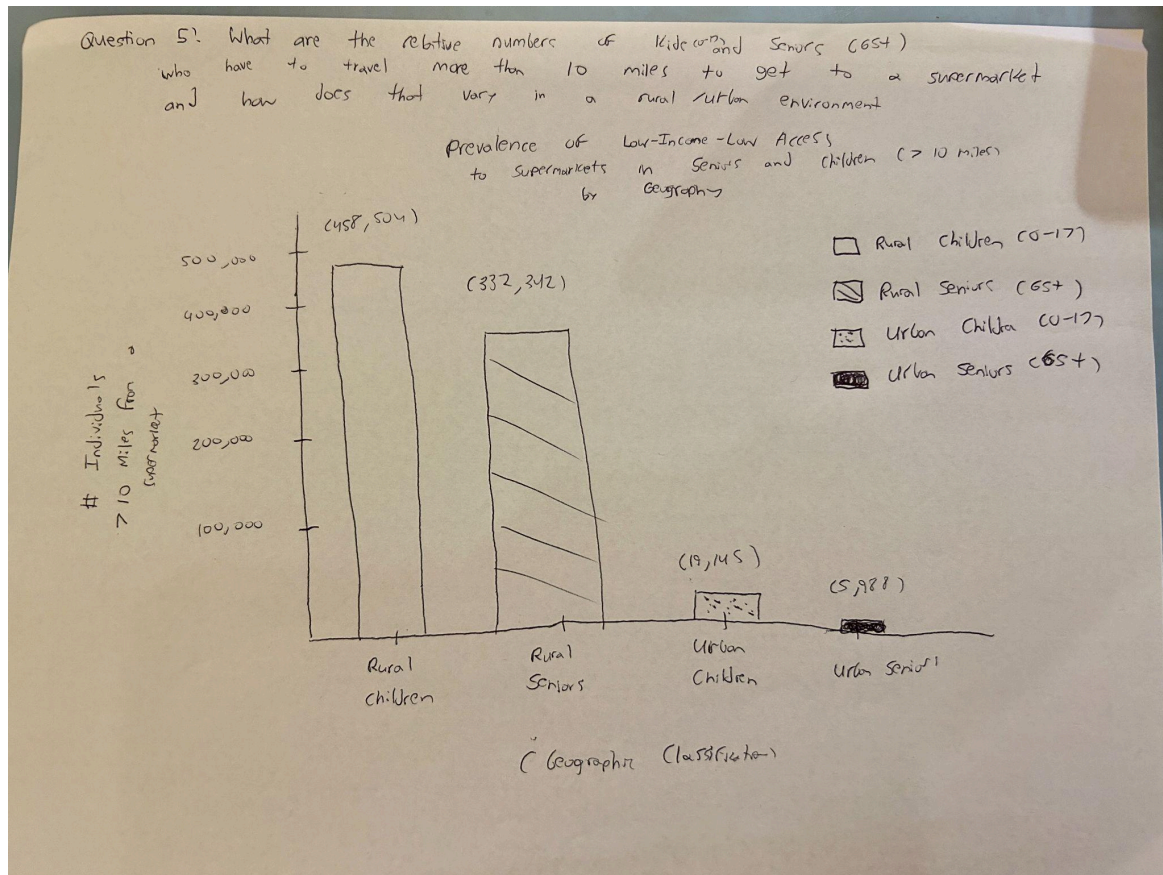
Sketch 4:



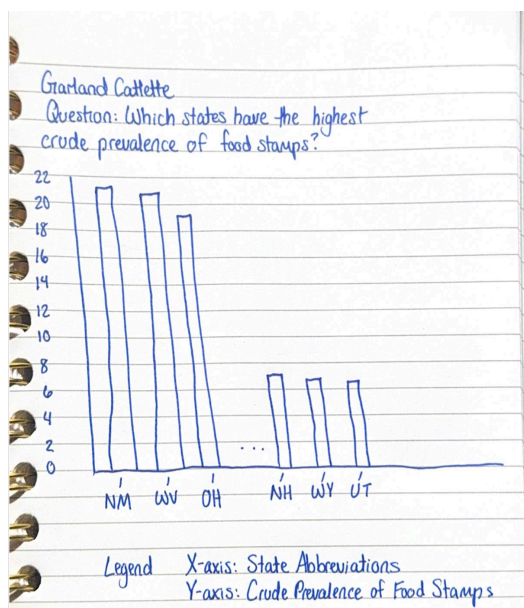
Sketch 12:



Sketch 5:



Sketch 10:



Reflection

We chose to implement these sketches because they provide a full picture on food deserts in the U.S., showing both geographic and demographic variations. Sketches 1/11 and 7/15 visualize the regional and state-level prevalence of food deserts which will allow for spatial trends in food access to be very clear. Sketches 4 and 5 highlight disparities by race, age, and rurality which will show a unique view on the social and structural factors contributing to food access inequities. Sketch 12 is an example of a sketch that will connect these general trends to concrete health outcomes and Sketch 10 complements this by showing the connection of these trends to economic aspects. The sketches together show what we believe to be a complete view of our topic.

Milestone 8: Storyboard

Main Message Identification

Oscar

- Rural Children (0-17 Years of Age) and Rural Seniors (65+) are 24x more likely and 55x more likely to have to travel more than 10 miles to get to a Supermarket relative to Urban Children and Urban Seniors respectively
- Southern and Midwestern States (with the exclusion of California) have the highest prevalence of Food Desert (Low-Income, Low-Access) Tracts with 5 Southern States and 4 Midwestern States respectively
- Among the Top 10 States with the largest proportion of individuals reporting depression, 6 out of the 10 are Southern States

Dani

- Of the 72,531 Census Tracts in the US, 13% of Tracts are in a Food Desert (Low-Income Low-Access Area), 11% of these Food Desert Tracts are from Urban Tracts and the other 2% are Rural Tracts
- Food desert prevalence is strongly linked to poverty and health outcomes with strong correlations specifically between poverty rate, obesity rate, and diabetes rate
- Urban tracts show a higher proportion of low-access tracts ($\approx 14\%$ vs. 8%), but rural tracts experience higher obesity and diabetes rates. This implies that while urban residents may face more localized food inaccessibility, rural populations suffer more severe health consequences from those access limitations—possibly due to transportation barriers and fewer healthcare options.

Garland

- Mississippi and Alabama are both among the 5 states with the highest food insecurity crude prevalence, diabetes crude prevalence, high blood pressure crude prevalence, and obesity crude prevalence.
- Mississippi, Texas, New Mexico, Alabama, and Georgia are the states with the highest average food insecurity crude prevalence of 23.03, 21.61, 20.88, 20.08, and 18.3 respectively.

Final Main Insight: Food deserts reveal an overlapping geography of poverty, poor health, and mental distress that is concentrated in the Rural South and Midwest.

Digital Storyboard

[Figjam](#)

Hook

Distance Slider (1, 10, 20) miles to a supermarket showing the amount of people in both rural and urban census tracts that have to travel far distances

Oscar Roscelli

Rising Insights

- Rural & Urban Prevalence of Kids/ Seniors in Food Deserts

Oscar Roscelli

- contrasting side-by-side charts of different regions in the US

Dani Elashm-Ortuf

- deep-dive by county for states with the highest correlations between food desert prevalence and health outcomes

Dani Elashm-Ortuf

Correlation between food insecurity and prevalence of chronic conditions like obesity, hypertension, and diabetes.

Gordon Catlett

Geographic Choropleth map indicating states with highest overlap in high poverty, depression, and food desert tracts

Oscar Roscelli

Main Message

- overall map of food deserts in the US; the data will show more severe food desert prevalence in the South & Midwest

Dani Elashm-Ortuf

- call-out boxes on certain states with particularly severe health outcomes

Oscar Roscelli

Solution

- Supporting local food production through urban gardens, community farms, and regional agriculture reduces transportation costs and increases fresh food availability. Improving food distribution networks and reducing food waste helps redirect surplus from retailers and restaurants to those in need. Investing in infrastructure for food storage and transportation is particularly important in rural regions.

Gordon Catlett

- Emergency food assistance through food banks, pantries, and meal programs provides critical short-term support. School meal programs (breakfast and lunch) help ensure children have consistent access to nutrition. Direct food assistance programs like SNAP (food stamps) help low-income families afford groceries

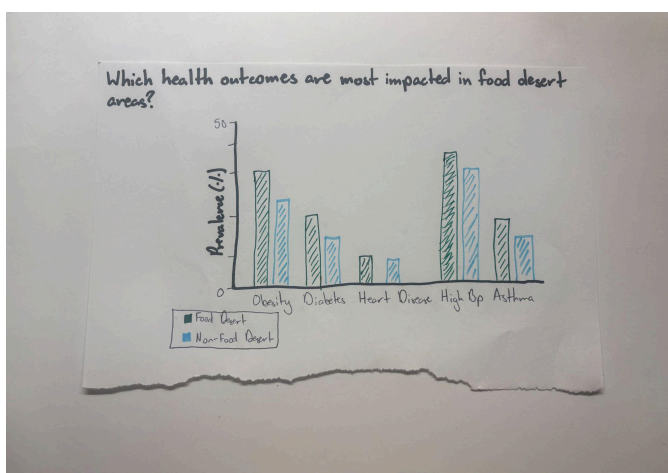
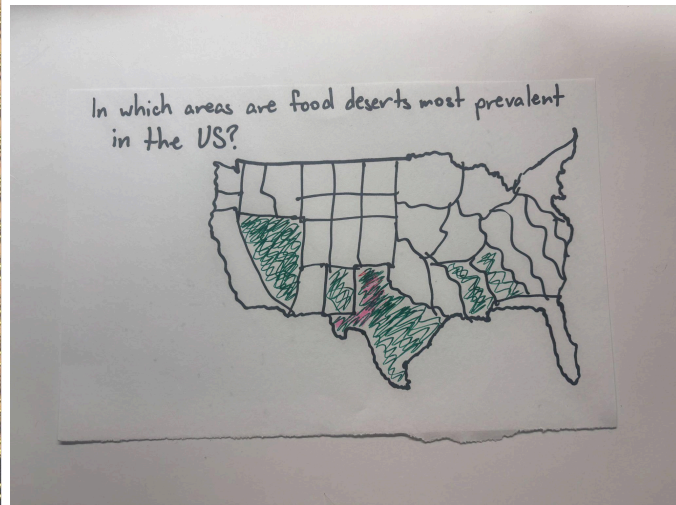
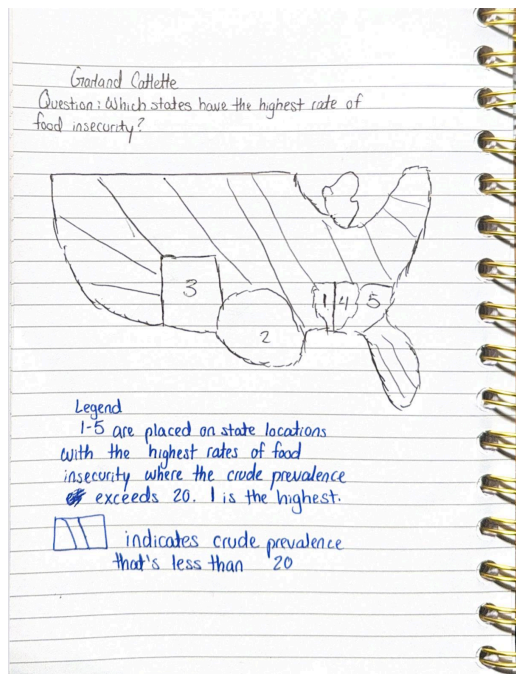
Gordon Catlett

Milestone 9: Prototype 1

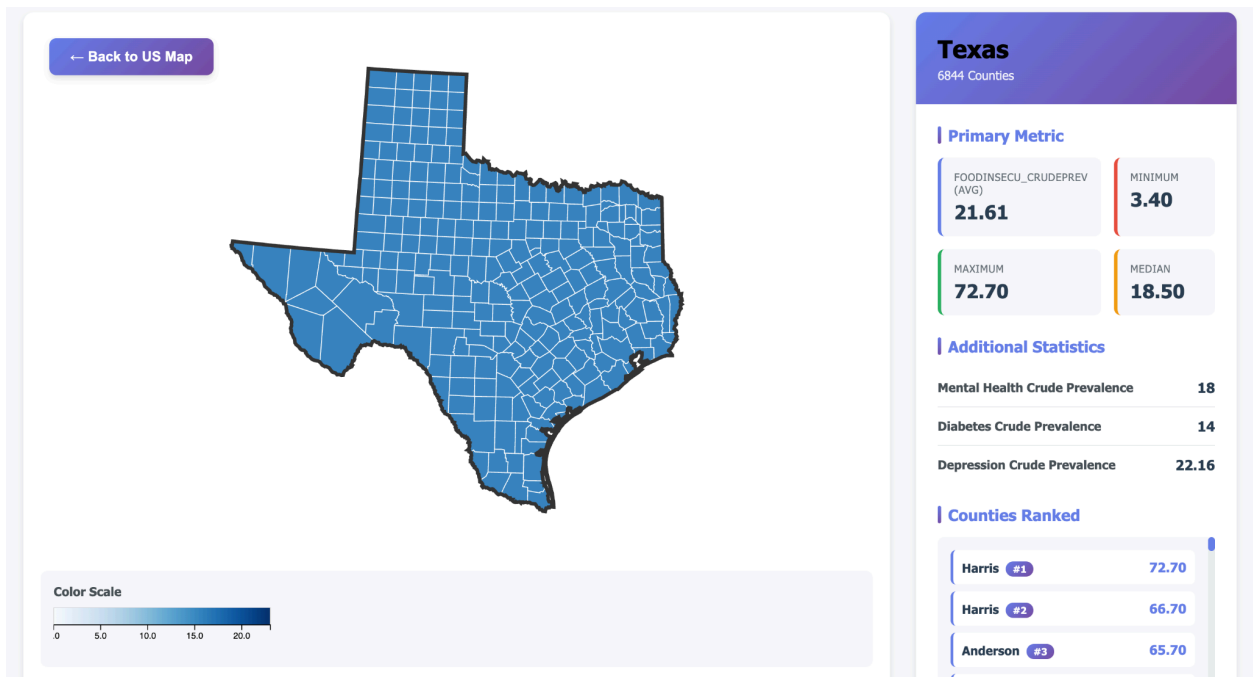
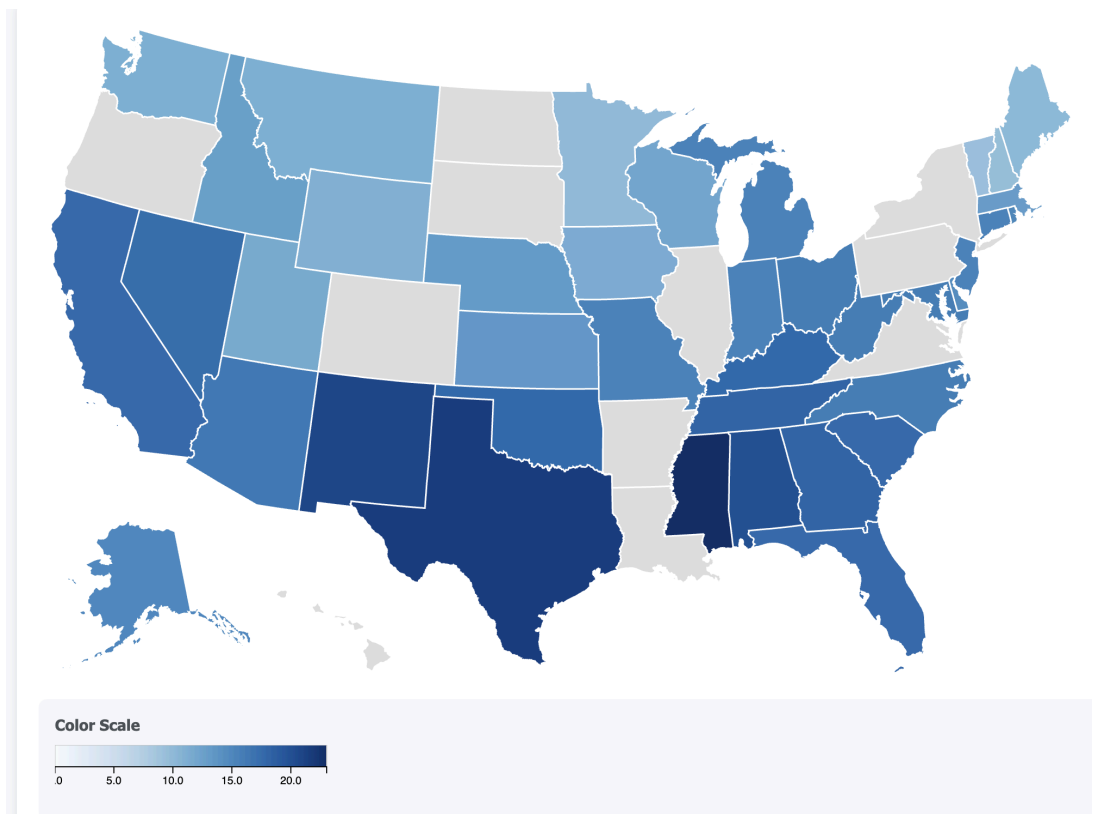
Food deserts reveal an overlapping geography of poverty, poor health, and mental distress that is concentrated in the Rural South and Midwest.

Two D3 Visualizations

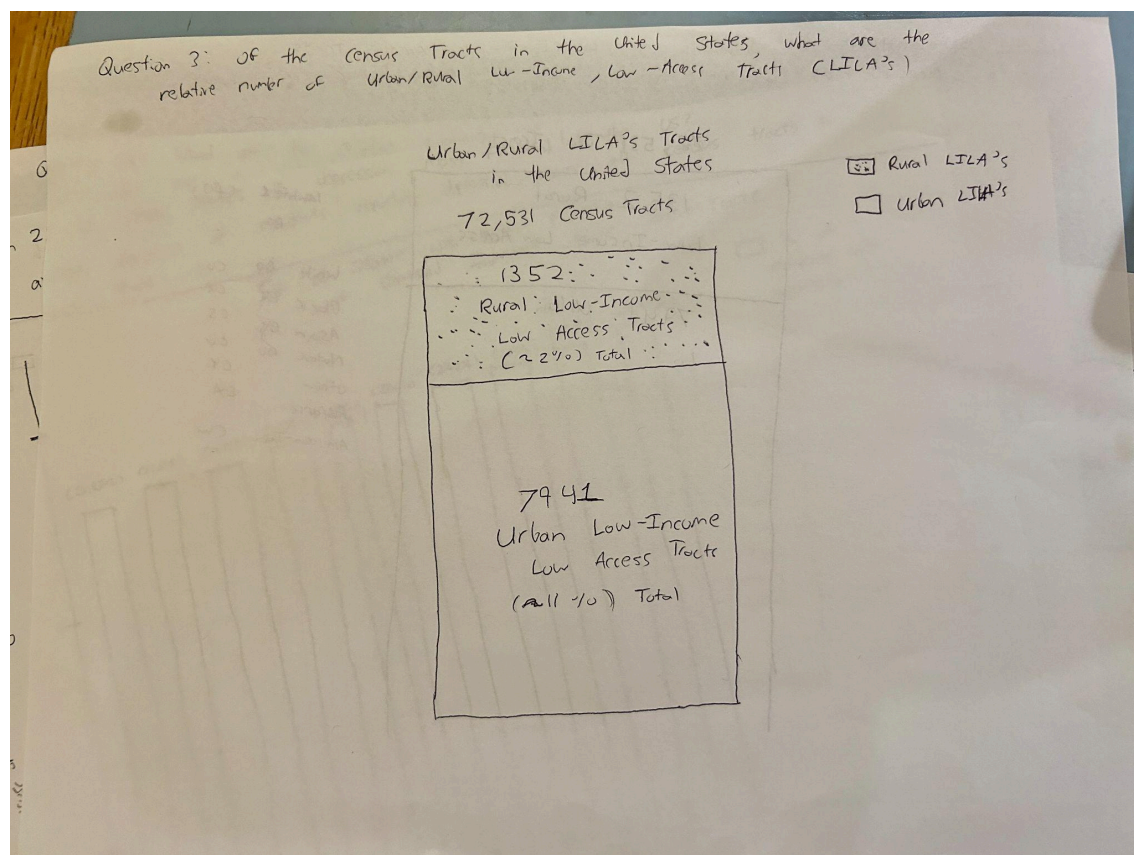
Sketch 7/15 (Combined with Sketch 12, Map Visualization):



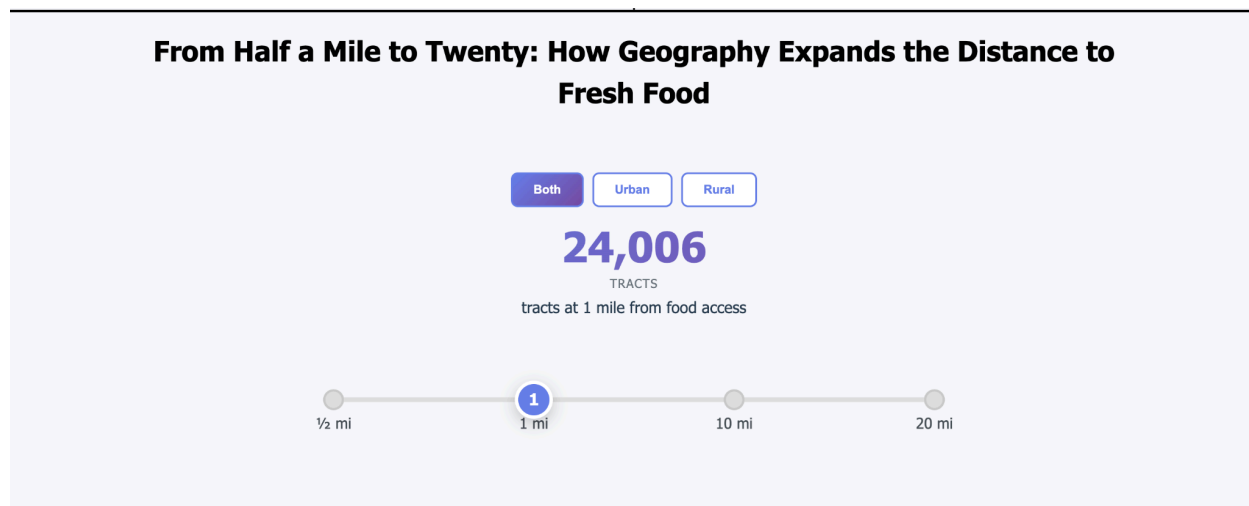
Visualization:



Sketch 3: Slider (conducted further data analysis resulting in adjusted numbers, accounted for rural/urban food desert definition distinctions)

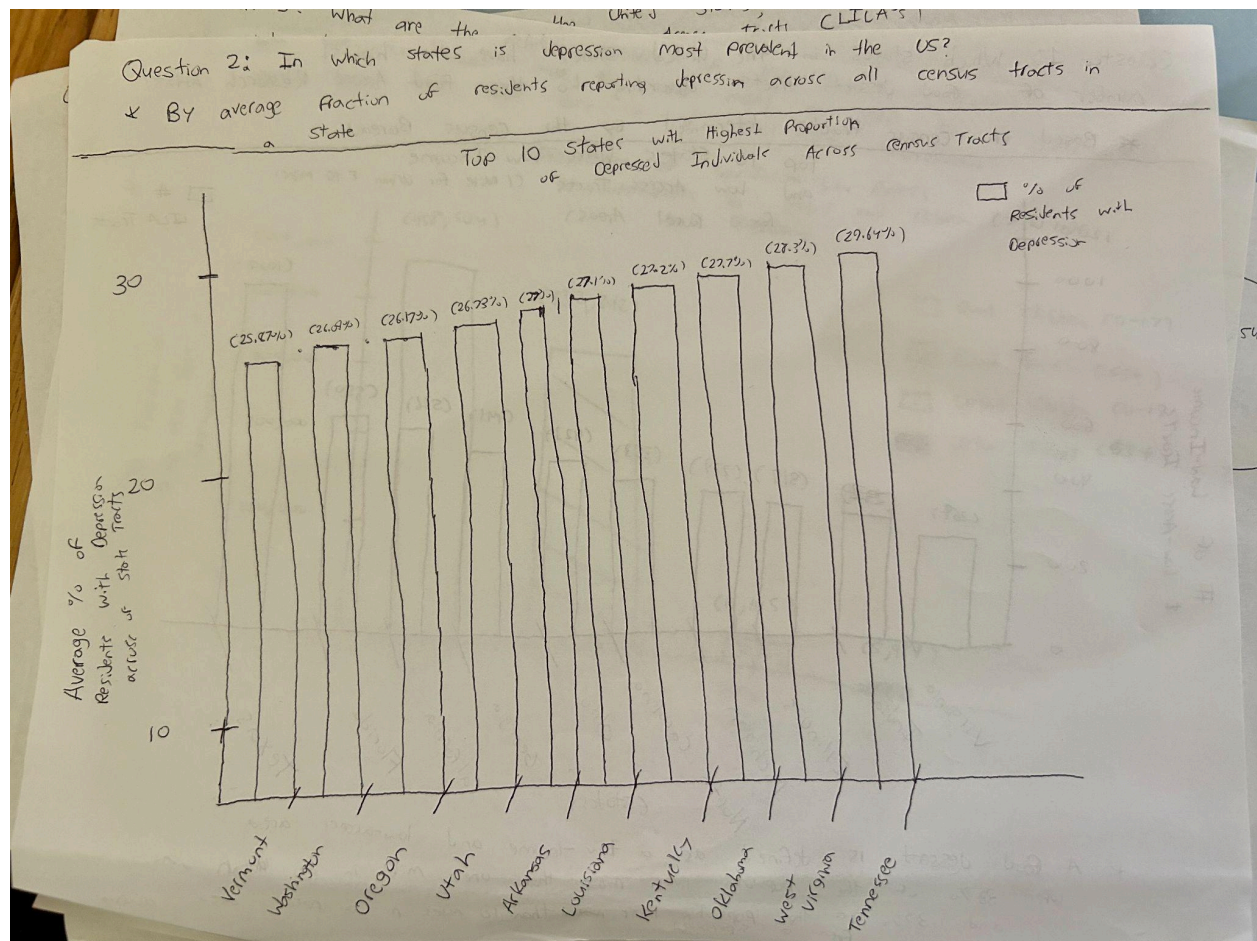


Visualization:



Additional Visualization Drafts:

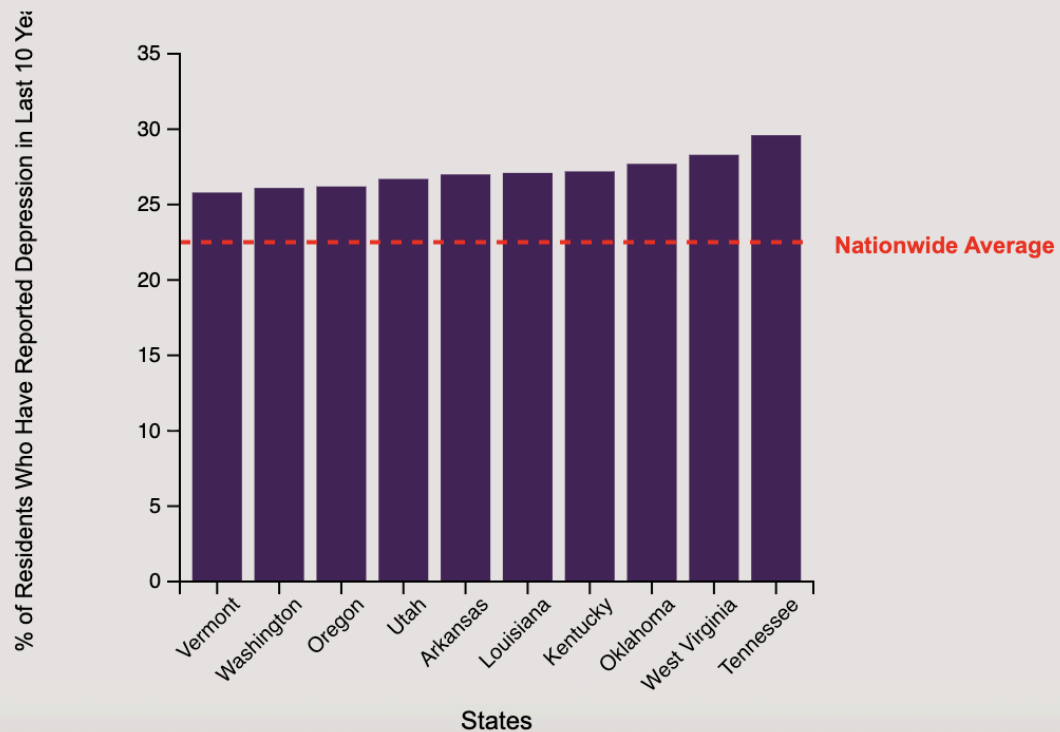
Sketch 2 (Food Desert Regional Distribution):



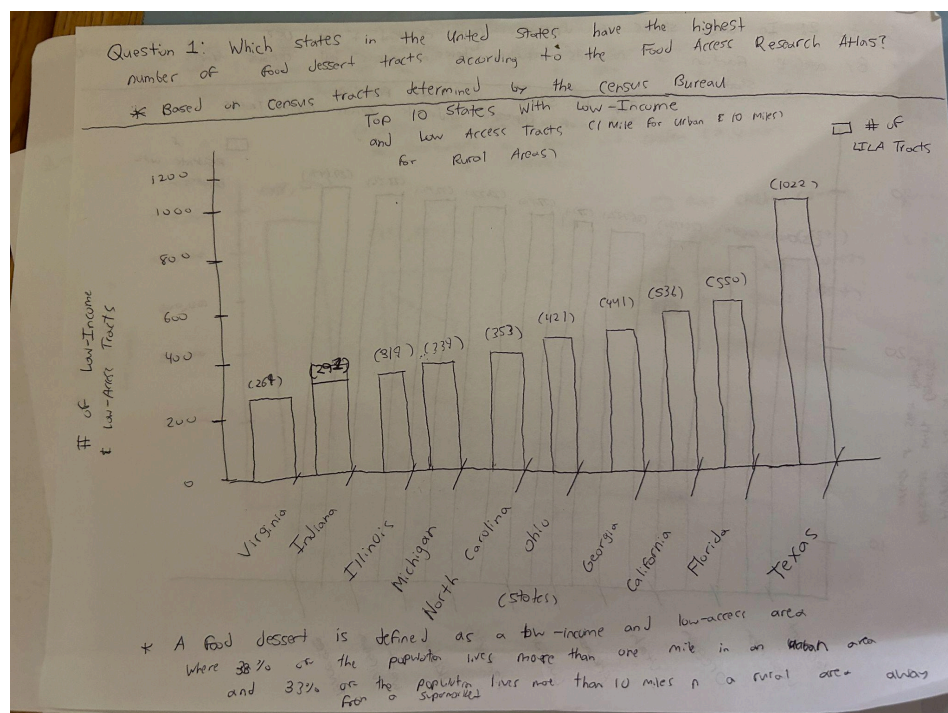
Implementation:

The Depression Landscape: America's Most Vulnerable States

Top 10 States with Highest Proportion of Depressed Individuals



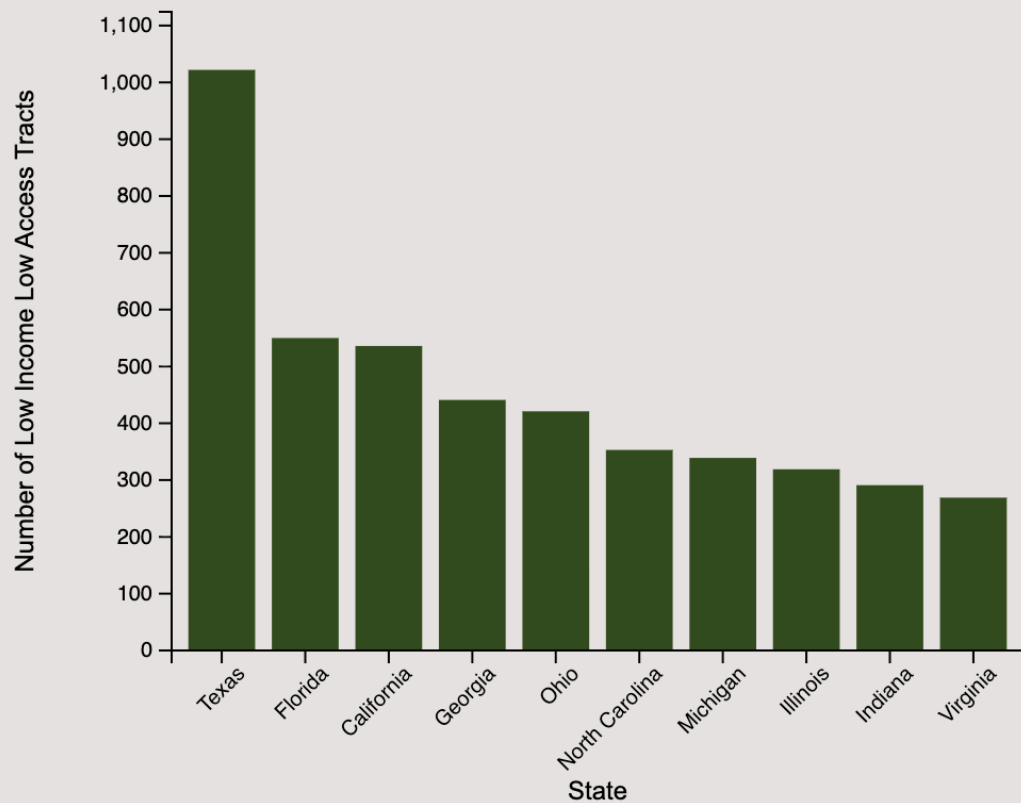
Sketch 1 (Top 10 States with Low Income Low Access Tracts):



Implementation:

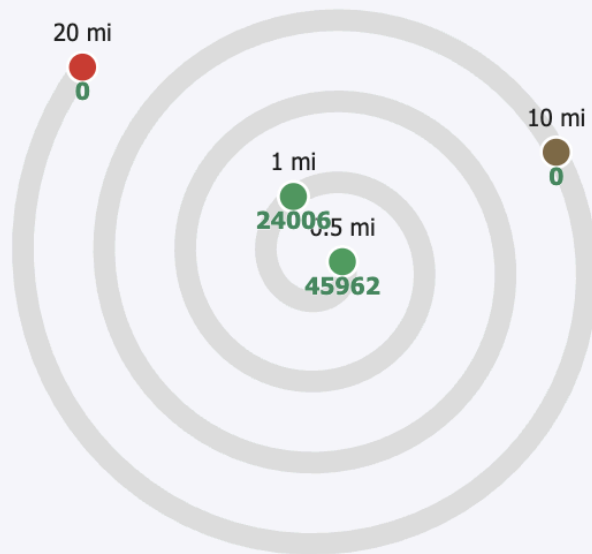
The Epicenters of Food Insecurity

Top 10 States with the Highest Number of Low-Income Low Access Tracts



Innovative Visualization (Roadmap):

Living on What's Left: The Everyday Reality of Food Deserts



20 mi



20 miles

Typical available food types:

- Processed snacks
- Soda
- Canned soups
- Fast food options

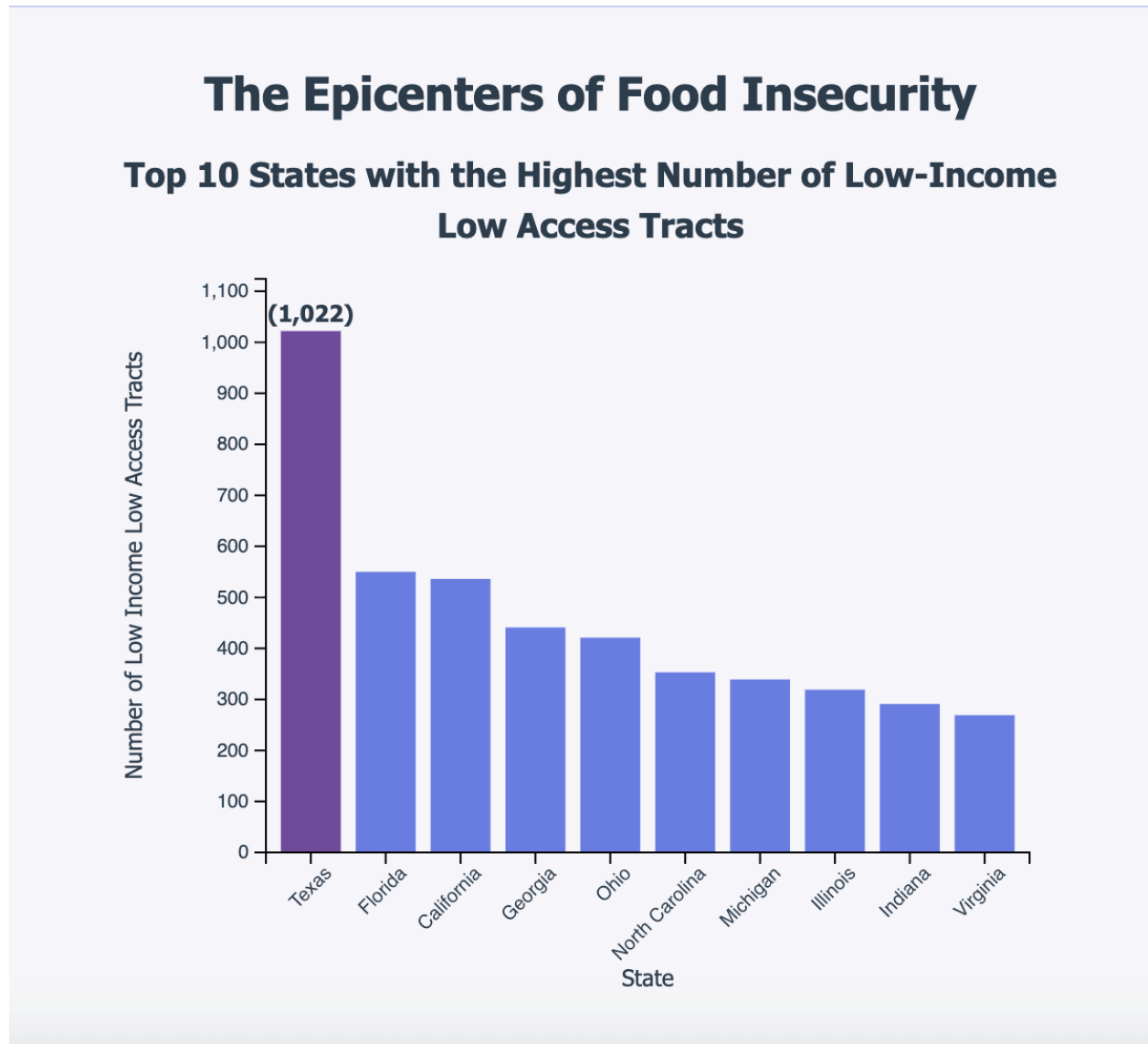
10 mi



Milestone 10: Prototype 2

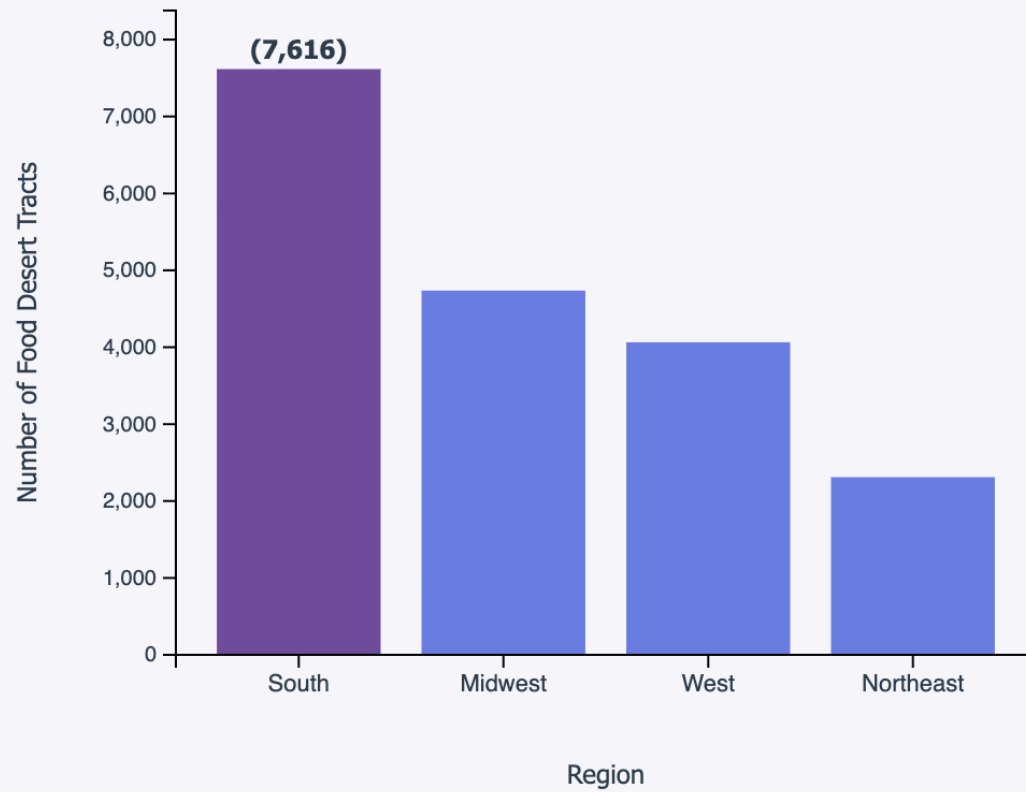
*We employed a new blue/purple theme for the website layout

First Changes (All Bar Graphs now D3)



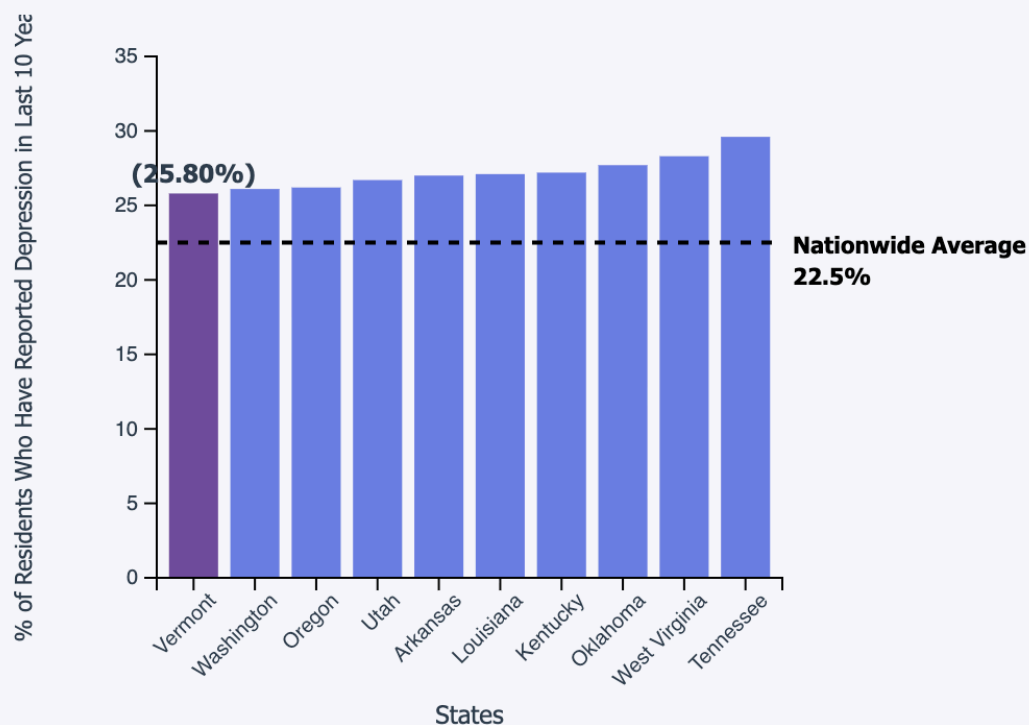
The Regional Prevalence of Scarcity

Regional Distribution of Food Desert Tracts



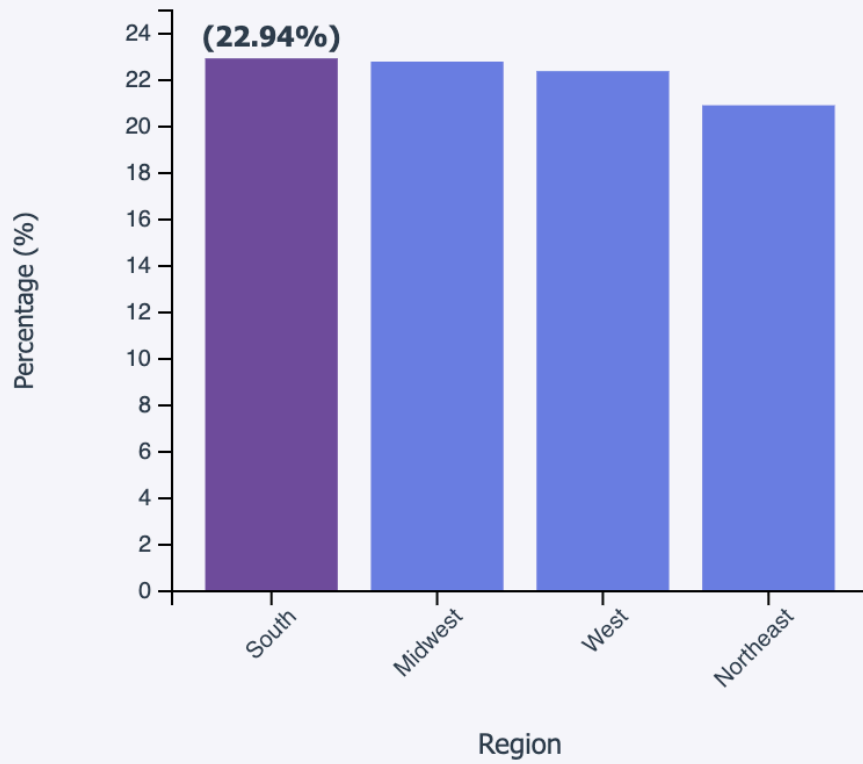
The Depression Landscape: America's Most Vulnerable States

Top 10 States with Highest Proportion of Depressed Individuals



Regional Patterns of Strain

Regional Distribution of Depression Prevalence



Updated Spiral/Slider Numbers

Both

Urban

Rural

46,335

TRACTS

tracts at 1/2 mile from food access

1/2 mi

1 mi

10 mi

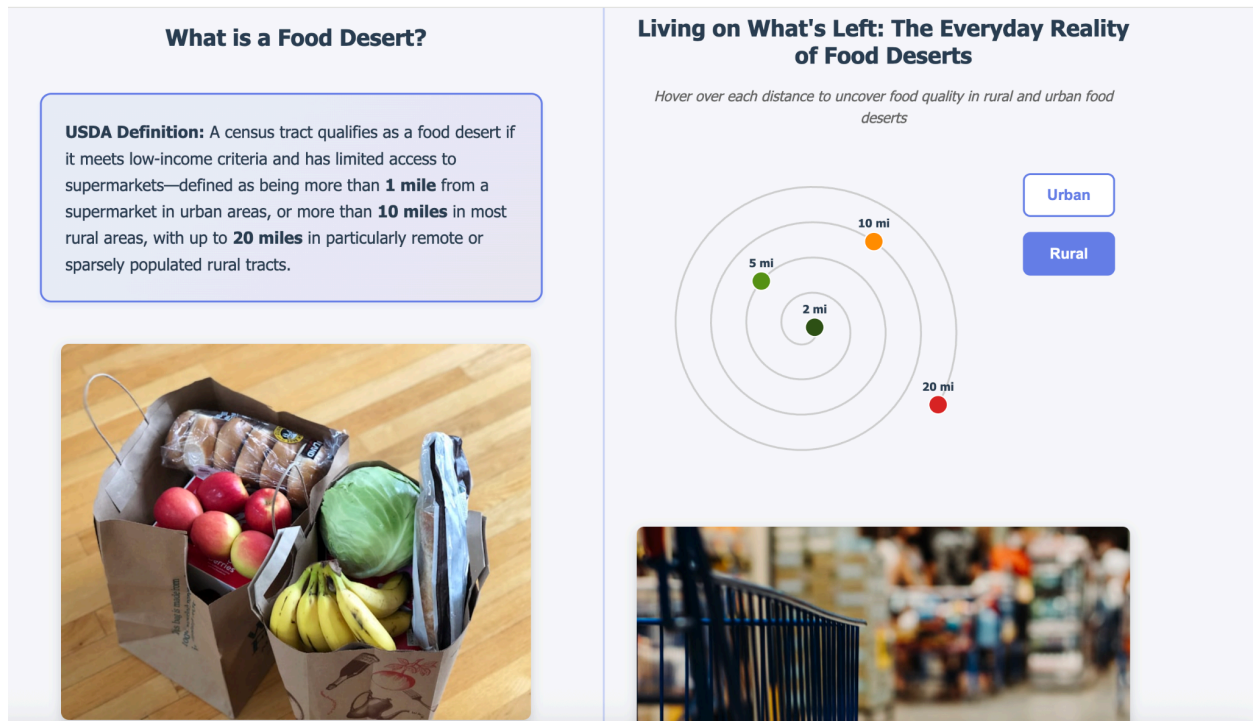
20 mi

Living on What's Left: The Everyday Reality of Food Deserts



Milestone 11: Prototype 3

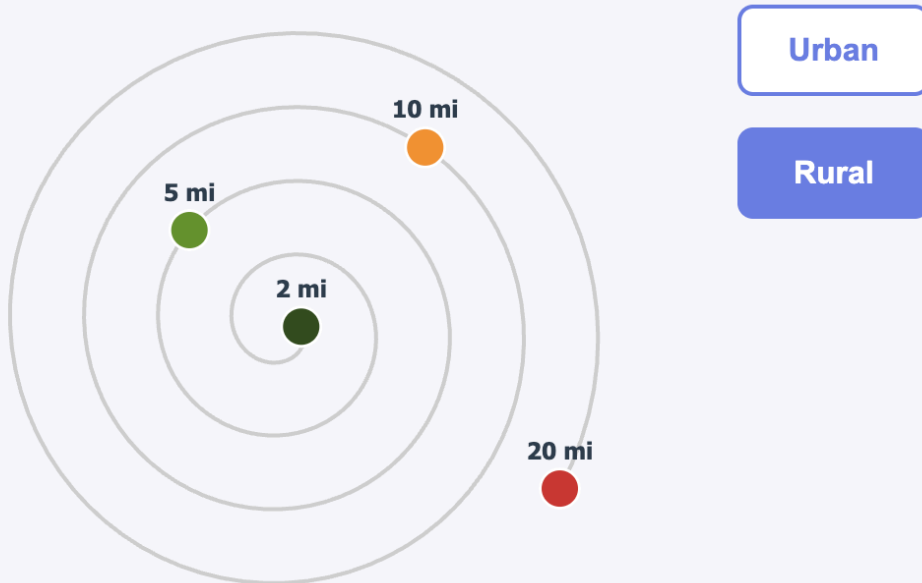
Updated Narrative: Included Images to add to narrative/theme of Food Deserts



Innovative Visualization (Introduced toggle on spiral, updated distances, introduced hover functionality on each distance to convey how food nutrition decreases with distance):

Living on What's Left: The Everyday Reality of Food Deserts

Hover over each distance to uncover food quality in rural and urban food deserts



USDA Definition: A census tract qualifies as a food desert if it meets low-income criteria and has limited access to supermarkets—defined as being more than 1 mile from a supermarket in urban areas, or more than 10 miles in rural areas.

Food options 2 miles away



What is a Food Desert?

USDA Definition: A census tract is considered a food desert if it meets low-income criteria and has limited access to healthy food.

Food options 20 miles away

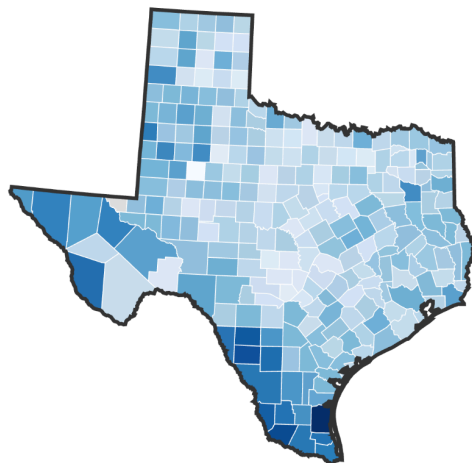


Living on what's Left: The everyday reality of Food Deserts

Hover over each distance to uncover food quality in rural and urban food deserts.

Additional Update (Map Visualization): Provided more context on numbers/metrics in sidebar, integrated health data that adjusts the map depending on the selected health metric), provides county level distributions

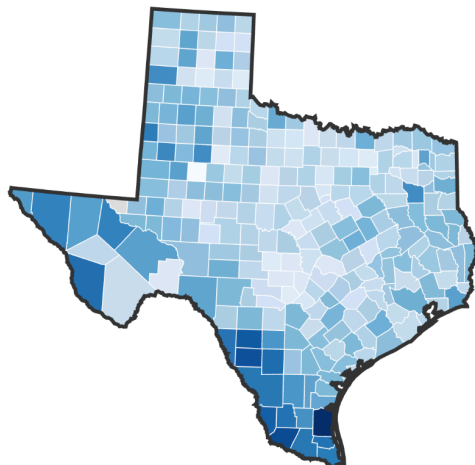
← Back to US Map



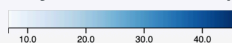
Unadjusted Rate of Food Insecurity (%)



← Back to US Map



Unadjusted Rate of Food Insecurity (%)



Texas

253 Counties

Select Health Metric:

Unadjusted Rate of Food Insecurity (%)

Selected Metric Statistics

AVERAGE

20.63

MINIMUM

6.70

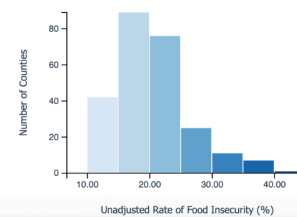
MAXIMUM

45.30

MEDIAN

19.66

County Distribution



Texas

253 Counties

Select Health Metric:

- ✓ Unadjusted Rate of Food Insecurity (%)
- Unadjusted Rate of Diabetes (%)
- Unadjusted Rate of Depression (%)
- Unadjusted Rate of Mental Illness (%)

AVERAGE

20.63

MINIMUM

6.70

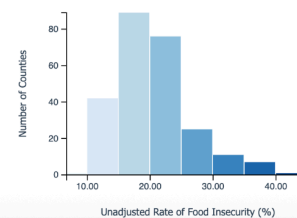
MAXIMUM

45.30

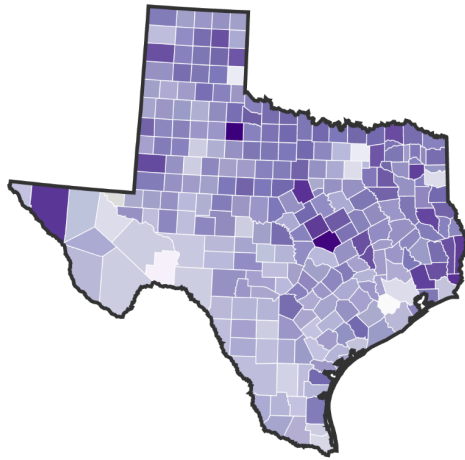
MEDIAN

19.66

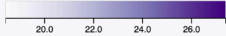
County Distribution



[← Back to US Map](#)



Unadjusted Rate of Depression (%)



Texas

253 Counties

Select Health Metric:

Unadjusted Rate of Depression (%)

Selected Metric Statistics

AVERAGE

23.15

MINIMUM

18.41

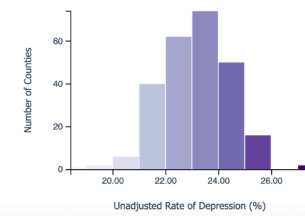
MAXIMUM

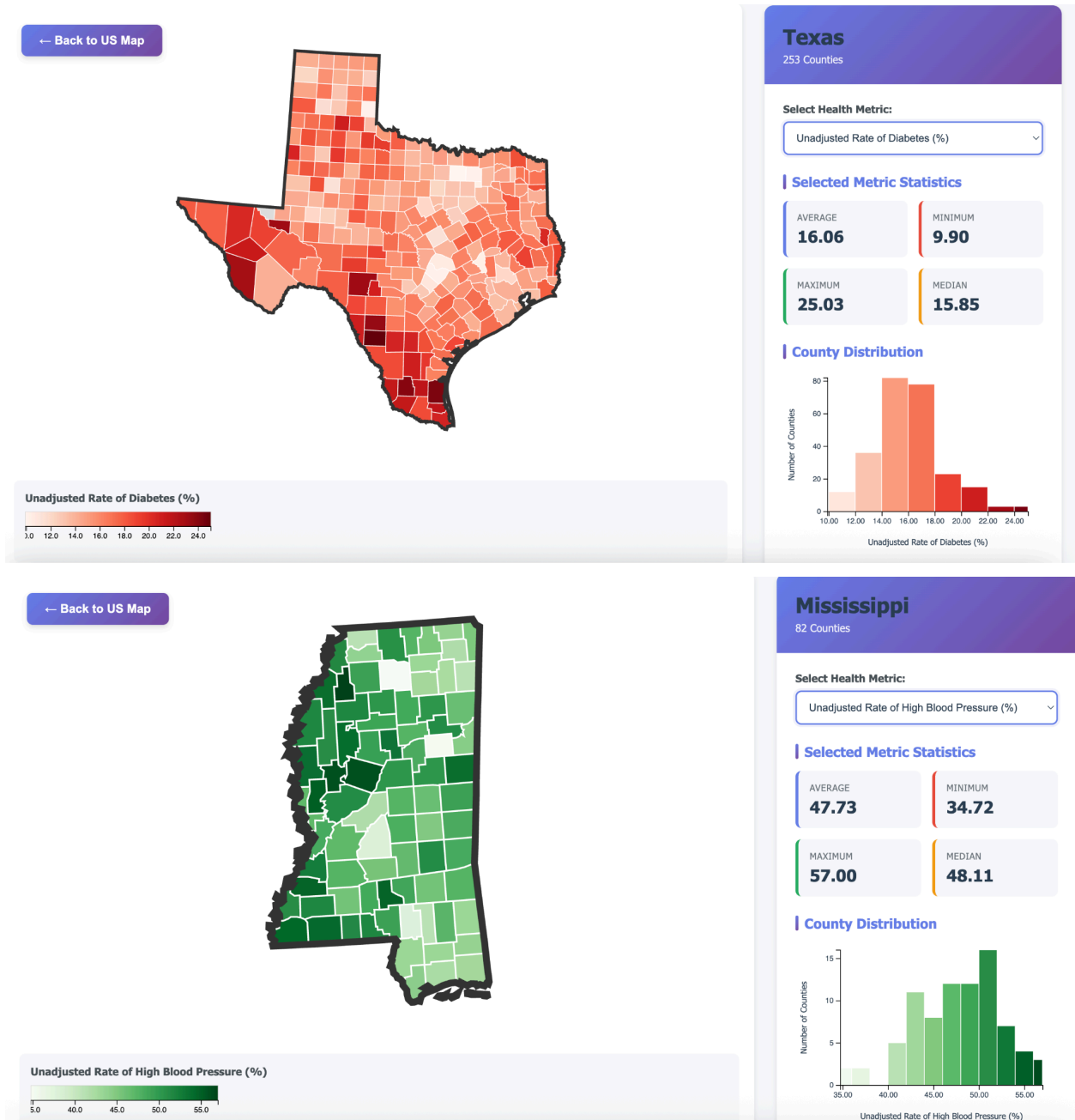
27.40

MEDIAN

23.13

County Distribution





← Back to US Map

Unadjusted Rate of High Blood Pressure (%)

35.0 40.0 45.0 50.0 55.0

Mississippi

82 Counties

Select Health Metric:
Unadjusted Rate of High Blood Pressure (%)

Selected Metric Statistics

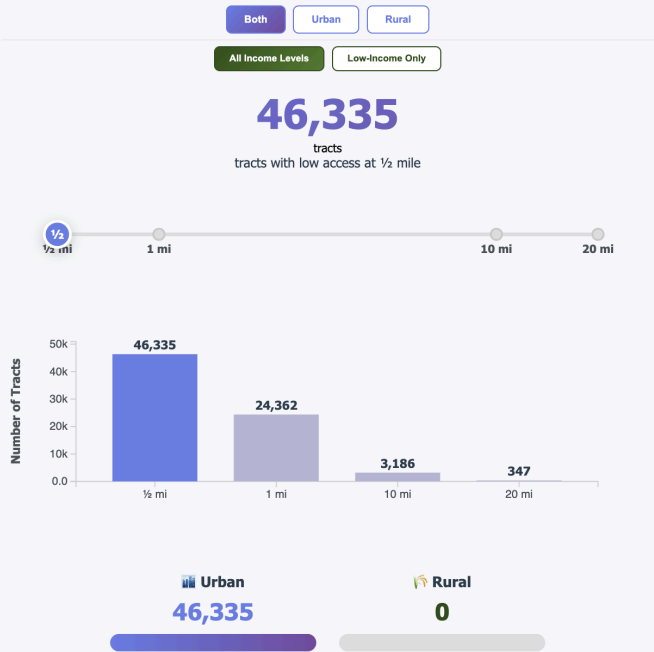
AVERAGE	47.73	MINIMUM	34.72
MAXIMUM	57.00	MEDIAN	48.11

County Distribution

Unadjusted Rate of High Blood Pressure (%)	Number of Counties
35.00-40.00	2
40.00-45.00	5
45.00-50.00	11
50.00-55.00	8
55.00-60.00	12
60.00-65.00	15
65.00-70.00	7
70.00-75.00	4
75.00-80.00	3

Updated Distance Slider to incorporate low-income distinction between tracts, logarithmic scale for clarity, and updated distances for Urban & Rural toggles to match with USDA definition

From Half a Mile to Twenty: How Geography Expands the Distance to Fresh Food



From Half a Mile to Twenty: How Geography Expands the Distance to Fresh Food



From Half a Mile to Twenty: How Geography Expands the Distance to Fresh Food

Both

Urban

Rural

All Income Levels

Low-Income Only

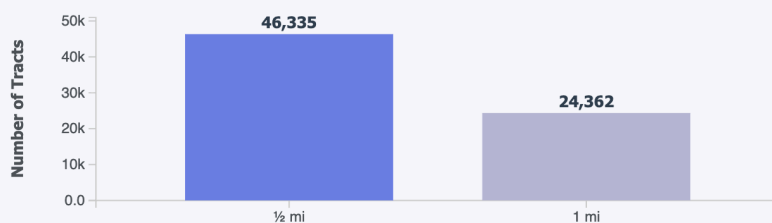
46,335

tracts

urban tracts with low access at ½ mile

½
½ mi

1 mi



From Half a Mile to Twenty: How Geography Expands the Distance to Fresh Food

Both

Urban

Rural

All Income Levels

Low-Income Only

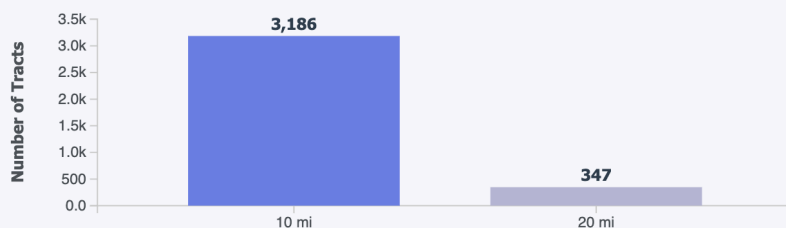
3,186

tracts

rural tracts with low access at 10 miles

10
10 mi

20 mi



Milestone 12: Think Aloud Results

	Notes (To be filled by project leads)
Tester Name #1	Kirthi Chigurupati kchigurupati@college.harvard.edu
Describe any usability issues or confusion the tester encountered while using the prototype.	Was confused by the dynamic bar below the slider, was confused about how the bar was filling up/tied to the slider being updated because the slider takes a lot of vertical spacing as opposed to horizontal spacing
Was the tester able to understand the main message of the data story? (e.g., Yes/No + why/why not?)	Didn't feel that the main message about the intersection of mental health and food deserts came through, says that a lot of it has to do with the order in which visualizations are presented
What parts of the interface or visualization did the tester find most engaging or effective?	- Enjoyed the slider and how it is dynamic - Was a huge fan of the map of the US, found it to be the strongest visualization, encouraged to push it up before and have it be the focal point of the narrative
What parts did the tester find confusing or less effective?	- Was confused by the messaging/narrative behind the spiral, asked "Does 0.1 miles mean this is what is available at that range?" - Forgot that tracts referred to census tracts, encouraged us to put the word "Census" anywhere where it says census tracts
Did the tester encounter any inconsistencies in design, data, or narrative?	Felt that the design data and narrative were consistent but felt that the main message didn't come through
Were there any unexpected interactions or insights that emerged during the session?	Voronoi diagram may not be the best for conveying community resources, seems to communicate a hierarchy to user of strength of community resources when there is no intended hierarchy, said displaying logos in circles would be the best
What specific improvements or changes did the tester suggest for the prototype?	Said the spiral visualization is great, but lacks substance in contributing to narrative, recommends making it more tangible by having the spiral display restaurant options in food desserts in both rural and urban environments

Did the tester suggest any additional insights or visualizations to include?	Said that all the content is there, thinks the only improvements have to do with the manner in which the content is presented, e.g. embed bar graphs into the map
General observations or comments from the tester.	Really like our visualizations on aggregate, was a fan of how we structured the narrative, and thinks that food deserts are a very important topic that deserve more attention

	Notes (To be filled by project leads)
Tester Name #2	Winifred Ofori-Manu
Describe any usability issues or confusion the tester encountered while using the prototype.	Thinks the visualizations would benefit from more context, e.g. placing captions/indicating what the functionality of each visualization is would be very helpful,
Was the tester able to understand the main message of the data story? (e.g., Yes/No + why/why not?)	Yes, felt that message about food desert and its associated negative outcomes came through, feels the headers do a good job of communicating the narrative effectively
What parts of the interface or visualization did the tester find most engaging or effective?	Huge fan of the Map of the United States, felt that it was incredibly engaging and kept playing around with it and experimenting with all of its functionality
What parts did the tester find confusing or less effective?	Was confused by the Urban/Rural bar at the bottom of the slider visualization
Did the tester encounter any inconsistencies in design, data, or narrative?	Felt that the design, and data were consistent throughout, thought the bar graphs with Top 10 States/Regional Distribution would be best after the map instead of coming before, was confused by the messaging behind the Regional Distribution of Depression prevalence, was confused as to whether it was intending to communicate that the distribution was equal
Were there any unexpected interactions or insights that emerged during the session?	Was taken aback by the spiral visualization, randomly discovered the hovering functionality
What specific improvements or changes did the tester suggest for the prototype?	Finds the split view of visualizations overwhelming, think it places a lot of cognitive load on the user, encouraged a carousel journey

	Also thought there was a hierarchy in the Community Organizations because of the Voronoi Diagram, encouraged the representation to be equal
Did the tester suggest any additional insights or visualizations to include?	Felt that there was no other visualizations to include, think all the content is there but that the way content is conveyed could be better in terms of the narrative
General observations or comments from the tester.	Likes what we've done so far, thinks there just needs to be a few more small changes to get it to a final stage

Think Aloud Discussion:

- Based on the results of your 'think aloud' study, what would you improve in your data story?
 - As it stands our Spiral visualization needs slightly stronger narrative, to make it more substantive, we'd like to make it so that the Spiral reflects food options in both an Urban and Rural food desert to make it more tangible to the user. Similarly, the Spiral's hover functionality is seen as jarring, therefore we'd like to make it so that the hover functionality opens a sidebar with these food options instead of changing the screen entirely
 - To make the community resources more interpretable, we'd like to substitute away from a Voronoi visualization into circular logos to help communicate the resources better, as before, the voronoi was communicating an unintended hierarchy of food desert and mental health community organizations for both testers
 - From a narrative flow, we received the feedback that the split view for the visualizations is overwhelming. To that end, we'd like to pivot to a carousel format to make the narrative more digestible and easy to follow so users can determine the take aways more easily
 - While others navigated our webpage, users expressed a common sentiment regarding our bar graphs stating that they wished they saw them in a map. Then, they were pleased to see the map after scrolling through. So, we'd like to allow users to see the map sooner rather than later by embedding visualizations and insights from the graphs into the map. Lastly, one user expressed that they would like to be able to interact with the map on the national level to a similar extent at which users can interact with the map on the state level.

- Are there any additional insights and visualizations you would use? Would you amplify or change your message? Did your narrative work? Did the tester get your takeaways?
 - Based on our user interviews, we do not need to use any additional visualizations or insights. With that said, we feel that our narrative would benefit from a carousel representation to make the message clearer and have it be a lot more amplified. Users felt that our narrative was strong at its foundation but would benefit from restructuring the order of our graphs (placing the map before the bar graphs) to make the takeaways, e.g. adverse health effects and who is impacted geographically clearer to users.
- Decide as a team which of these improvements you will implement and write down your decisions and why you made them in your process book as a numbered list.
 1. Spiral Visualization Improvement (Baldwin Florida will be our Rural City Example, shows that fast food is easily accessible but supermarkets are >10 miles away, will construct a sidebar for visualization for the hovering functionality, Maxwell Park (Oakland) California will be our Urban example to convey a similar trend)
 2. Community Resources: We will adjust the Voronoi diagram to be circular clickable logos for all the community organizations
 3. Distance Slider Visualization Improvement: We will update the distance slider to take up less vertical real estate so that all of our intended takeaways are easily visible and understandable. We will also remove the bars below the distance slider as they are confusing and communicate the same information as above.
 4. Map Improvement: We will update the map by embedding an information panel on the national level of the map, allowing users to explore data and trends on a national level. Secondly, the placement of the map will be updated so that users can spend most of their time navigating most of our data that will be featured in the map.
- Implement the intended changes and check them off your list (e.g., adding “done”). You can distribute the tasks among your team members. If you are unable to implement specific changes, please explain why and describe the expected results in your process book.
 - Spiral Visualization Improvement
 - Done!
 - Community Resources Adjustment
 - Done!
 - Distance Slider Visualization Improvement
 - Done!
 - Map Improvement
 - Done!
 - Carousel Functionality
 - Done!